MOG JEE Main Full Test - 2024 Revised

Total Time: 180 minutes | Total Marks: 300 | Total Questions: 75

GENERAL INSTRUCTIONS

The Question Paper consists of three parts (Physics, Chemistry and Mathematics) with a total of **75** questions.
Each subject is divided into 2 sections:
Section 1: Single Correct Type Questions (4 options, only one correct)
Section 2: Numerical Value Type Questions (answer to be entered as numerical value)
Marking Scheme:
Single Correct: +4 for correct, -1 for wrong, 0 if unattempted.
Numerical Value: +4 for correct, -1 for wrong.

By: C I P H Ξ R

Q.1

Maximum kinetic energy of a photoelectron is E, when the wavelength of incident light is λ. If energy becomes 4 times, when wavelength is reduced to $\frac{1}{3}\lambda$, the work-function is

(A) $\frac{3hc}{\lambda}$

(B) $\frac{hc}{\lambda}$

(C) $\frac{hc}{2\lambda}$

(D) $\frac{hc}{3\lambda}$

Q.2

Assertion: Fundamental frequency of a narrow organ pipe is more than a broad diameter organ pipe of the same length.
Reason: If we apply Laplace end correction, then the fundamental frequency of a closed organ pipe is more than $\frac{v}{4l}$.

(A) Both assertion and reason are correct and reason is the correct explanation of assertion.

(B) Both assertion and reason are true but reason is not the correct explanation of assertion.

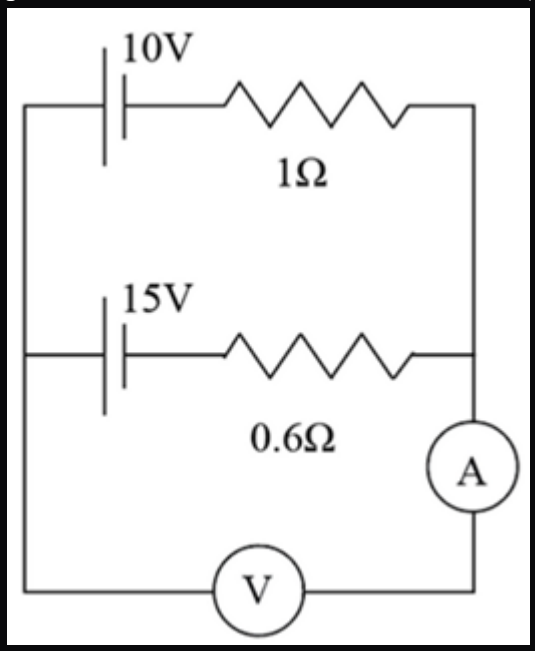
(C) Assertion is true but reason is false.

(D) Assertion is false but Reason is true.

Q.3

A $10\ \text{V}$ battery with internal resistance 1Ω and a $15\ \text{V}$ battery with internal resistance 0.6Ω are connected in

parallel to a voltmeter of resistance 1Ω (see figure). The readings in the ideal ammeter is :



(A) $\frac{115}{11}A$

(B) $\frac{105}{11}$ A

(C) $\frac{105}{21}$ A

(D) $\frac{115}{21}$ A

Q.4

A rod of length l and cross-section area A has a variable thermal conductivity given by $K = \alpha T$, where α is a positive constant and T is the temperature in Kelvin. Two ends of the rod are maintained at temperatures T_1 and T_2 $(T_1 > T_2)$. Heat current flowing through the rod will be -

(A) $\frac{A\alpha\left(T_1^2 - T_2^2\right)}{3l}$

(B) $\frac{A\alpha\left(T_1^2 + T_2^2\right)}{l}$

(C) $\frac{A\alpha\left(T_1^2 + T_2^2\right)}{3l}$

(D) $\frac{A\alpha\left(T_1^2 - T_2^2\right)}{2l}$

Q.5

When white light passes through a dispersive medium, it breaks up into various colours. Which of the following statements is true?

(A) Velocity of light for violet is greater than the velocity of light for red colour.

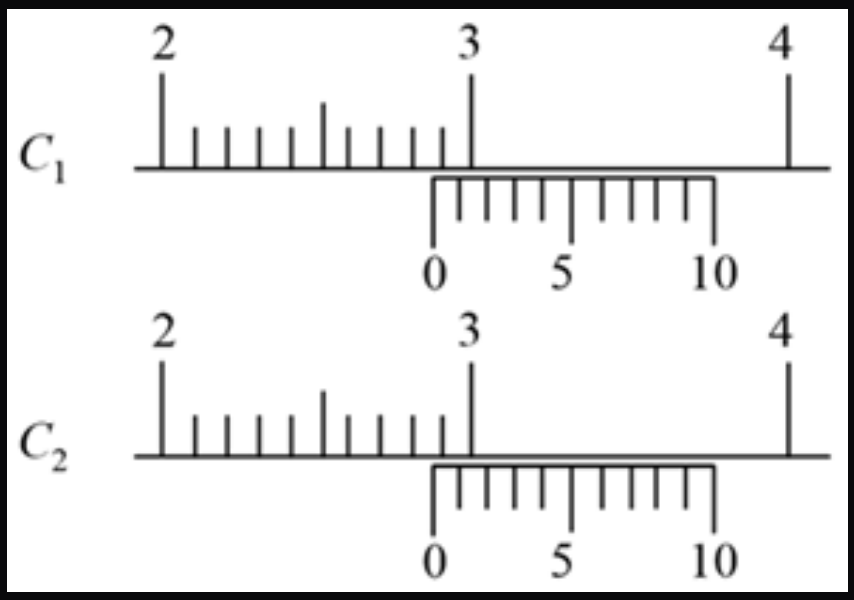
(B) Velocity of light for violet is less than the velocity of light for red.

(C) Velocity of light is the same for all colours

(D) Velocity of light for different colours has nothing to do with the phenomenon of dispersion

Q.6

There are two vernier calipers both of which have 1 cm divided into 10 equal divisions on the main scale. The Vernier scale of one of the calipers (C_1) has 10 equal divisions that correspond to 9 main scale divisions. The Vernier scale of the other caliper (C_2) has 10 equal divisions that correspond to 11 main scale divisions. The readings of the two calipers are shown in the figure. The measured values (in cm) by calipers C_1 and C_2 respectively, are (in both cases 7th Vernier Scale Division coincides with Main Scale Division)



(A) 2.87 and 2.86

(B) 2.87 and 2.82

(C) 2.87 and 2.83

(D) 2.85 and 2.82

Q.7

Electric displacement is given by $D = \varepsilon E$, here, ε =electric permittivity, E =electric field strength
The dimensions of electric displacement are

(A) $\left[\mathrm{ML}^{-2}\mathrm{TA}\right]$

(B) $\left[\mathrm{L}^{-2}\mathrm{T}^{-1}\mathrm{A}\right]$

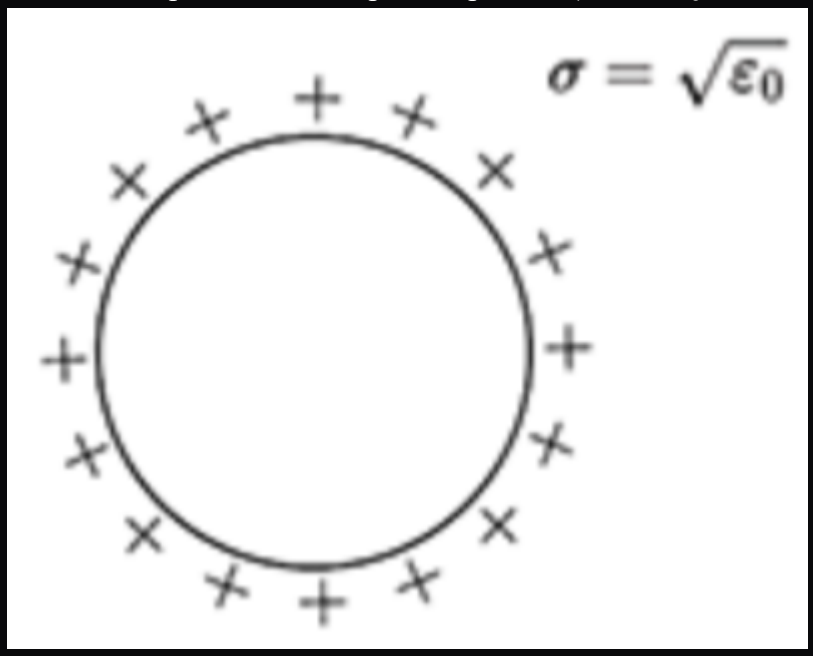
(C) $\left[\mathrm{L}^{-2}\mathrm{TA}\right]$

(D) None of these

Q.8

Charge is distributed over a surface of hollow conducting sphere of surface charge density $\sigma = \sqrt{\varepsilon_0}$, Then

electrostatic pressure developed in sphere is(where $\varepsilon_0 = 8.85 \times 10^{-12} \mathrm{Nm}^2 c^2$)



(A) $0.5N/m^2$

(B) $0.5\varepsilon_0 N/m^2$

(C) $\frac{2}{\varepsilon_0} N/m^2$

(D) $2.\mathrm{N}/m^2$

Q.9

For steel, the Young's modulus and the strain at the breaking point are $2 \times 10^{11}\ \mathrm{N\ m^{-2}}$ and 0.15, respectively. The stress at the break point for steel is

(A) $1.33 \times 10^{11}\ \mathrm{N\ m^{-2}}$

(B) $1.33 \times 10^{12}\ \mathrm{N\ m^{-2}}$

(C) $7.5 \times 10^{-3}\ \mathrm{N\ m^{-2}}$

(D) $3 \times 10^{10}\ \mathrm{N\ m^{-2}}$

Q.10

Hydrogen atoms absorbs radiation of wavelength λ_0 and consequently emit radiations of 6 different wavelengths of which two wavelengths are shorter than λ_0.

1. The final excited state of the atoms is $n = 4$
2. The initial state of the atoms may be $n = 2$
3. The initial state of the atoms may be $n = 3$
4. There are three transitions belonging to Lyman series.

Which of the above are correct?

(A) $1, 2, 3, 4$

(B) $1, 4$

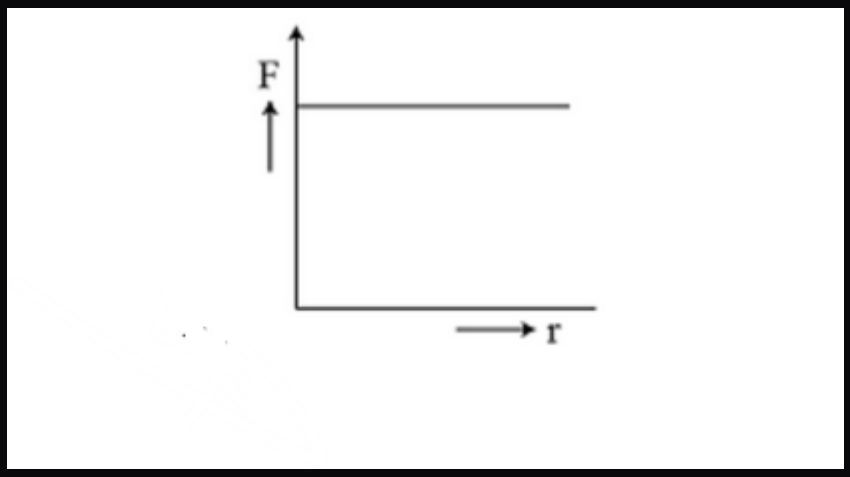
(C) $2, 4$

(D) $1, 2, 4$

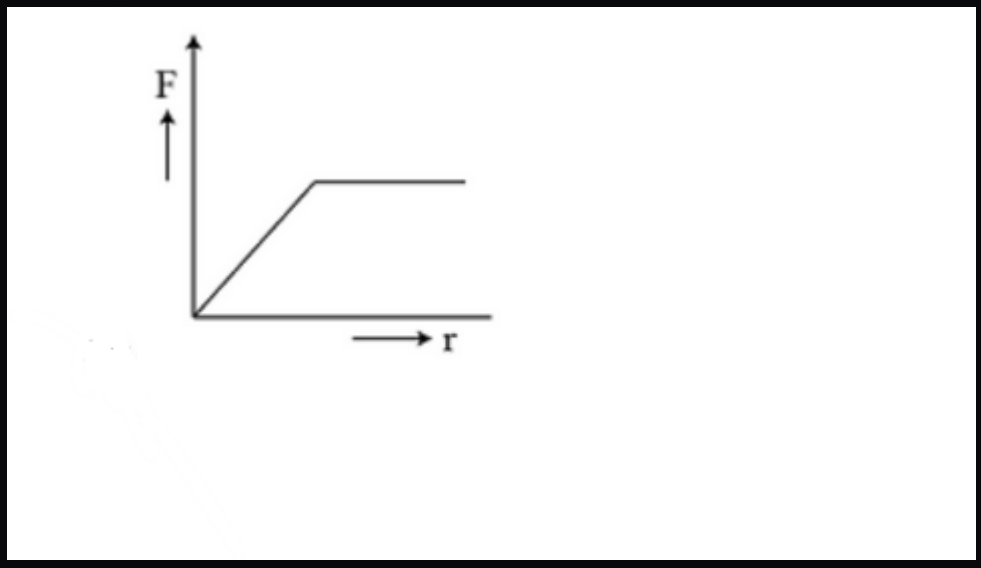
Q.11

If a particle moving in a circular path maintains a constant speed of $20\ \mathrm{ms}^{-1}$, then which of the following correctly describes relation between force (F) and radius (r) ?

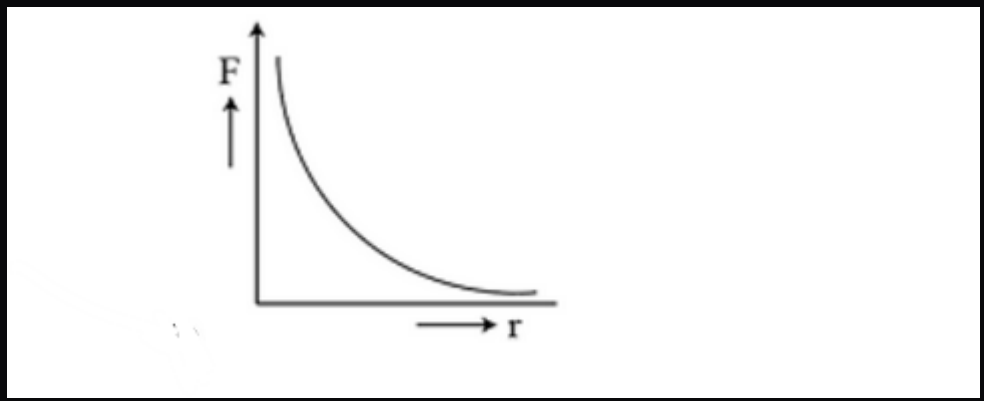
(A)



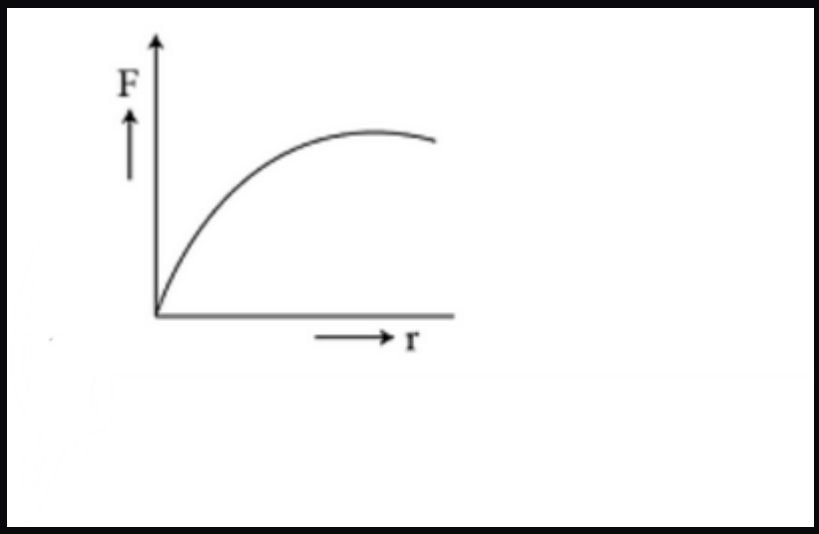
(B)



(C)

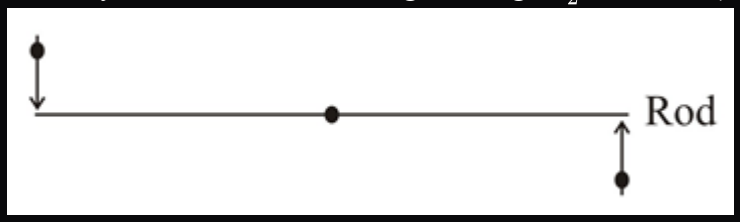


(D)



Q.12

A rigid uniform rod of mass M and length L is resting on a smooth horizontal table. It is pivoted at its centre. Two marbles each of mass $\frac{M}{6}$ moving with uniform speed L m s^{-1} in the plane of the table collide with the two ends of the rod simultaneously as shown in the figure. The marbles stuck to the rod and continue to move with the rod. Time taken by the rod to rotate through an angle $\frac{\pi}{2}$ radian is (in seconds)



(A) 1

(B) 2π

(C) π

(D) $\frac{\pi}{2}$

Q.13

A solenoid of 1.5 m length and 4.0 cm diameter has 10 turns per cm. A current of 5 A is flowing through it. The magnetic field induction at axis inside the solenoid is

(A) $2\pi \times 10^{-4}$ T

(B) $2\pi \times 10^{-5}$ T

(C) $2\pi \times 10^{-3}$ T

(D) $2\pi \times 10^{-6}$ T

Q.14

Let ω is the angular velocity of the earth's rotation about its own axis and acceleration due to gravity on the earth's surface at the poles is g. An object weighed by a spring balance gives the same reading at the equator, as well as at a height h above the poles $(h \ll R)$. What is the value of h ?

(A) $\frac{gR}{\omega}$

(B) $\frac{\omega^2 R^2}{g}$

(C) $\frac{\omega^2 R^2}{2g}$

(D) $\frac{2\omega^2 R^2}{g}$

Q.15

In a diode, the diffusion current is $i_{\text{diffusion}}$ and drift current is i_{drift}, then match the column.

Column-I	Column-II
(1) Diode with no biasing	(p) $i_{\text{diffusion}} >> i_{\text{drift}}$
(2) Diode in forward bias	(q) $i_{\text{drift}} >> i_{\text{diffusion}}$
(3) Diode in reverse bias without breakdown	(r) $i_{\text{drift}} > i_{\text{diffusion}}$
(4) Diode in reverse bias with breakdown	(s) $i_{\text{drift}} = i_{\text{diffusion}}$

(A) $A \to s, B \to q, C \to r, D \to p$

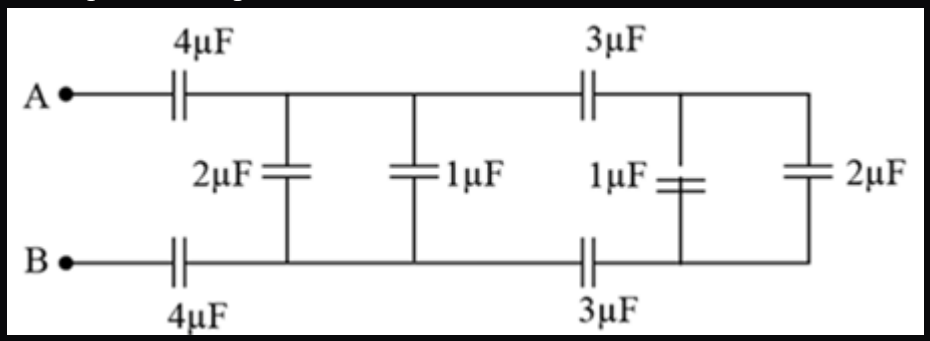
(B) $A \to s, B \to p, C \to r, D \to q$

(C) $A \to s, B \to q, C \to p, D \to r$

(D) $A \to q, B \to p, C \to r, D \to s$

Q.16

The equivalent capacitance between A and B is



(A) $\frac{2}{3}\mu F$

(B) $\frac{4}{3}\mu F$

(C) $\frac{3}{2}\mu F$

(D) $\frac{1}{2}\mu F$

Q.17

A simple harmonic motion is represented by $y = 5\left(\sin 3\pi t + \sqrt{3}\cos 3\pi t\right)$ cm. The amplitude and time period of the motion are

(A) $10 \text{ cm}, \frac{3}{2} \text{ s}$

(B) $5 \text{ cm}, \frac{2}{3} \text{ s}$

(C) $5\ \text{cm}, \frac{3}{2}\ \text{s}$

(D) $10\ \text{cm}, \frac{2}{3}\ \text{s}$

Q.18

In YDSE, the slits have different widths. As a result, the amplitude of waves from two slits is A and $2A$ respectively. If I_0 be the maximum intensity of the interference pattern then the intensity of the pattern at a point where the phase difference between waves is ϕ is given by $\frac{I_0}{P}(5+4\cos\phi)$. Where P is in integer. Find the value of P?

(A) 9

(B) 3

(C) 6

(D) 1

Q.19

A gas in container A is in thermal equilibrium with another gas in container B. Both contain equal masses of the two gases in the respective containers. Which of the following can be true?

(A) $\mathrm{P_A V_A = P_B V_B}$

(B) $\mathrm{P_A = P_B, V_A \neq V_B}$

(C) $\mathrm{P_A = P_B, V_A = V_B}$

(D) $\frac{\mathrm{P_A}}{\mathrm{V_A}} = \frac{\mathrm{P_B}}{\mathrm{V_B}}$

Q.20

Frequencies of various radiations are given as
$f_v \rightarrow$ Visible light
$f_r \rightarrow$ Radio waves
$f_{uv} \rightarrow$ Ultra Violet waves
Then which of the following is true?

(A) $f_{uv} < f_v < f_r$

(B) $f_r < f_v < f_{uv}$

(C) $f_v < f_r < f_{uv}$

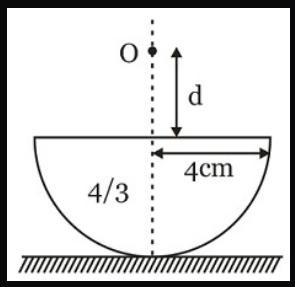
(D) $f_{uv} < f_r < f_v$

Q.21

In a series $L-C-R$ circuit, $C = 10^{-11}\ \mathrm{F}, L = 10^{-5}\mathrm{H}$ and $R = 100\ \Omega$, when a constant DC voltage E is applied to the circuit, the capacitor acquires a charge $10^{-9}\mathrm{C}$. The DC source is replaced by a sinusoidal voltage source in which the peak voltage E_0 is equal to the constant DC voltage E. At resonance the peak value of the charge acquired by the capacitor is $10^{-n}\mathrm{C}$. Mark value of n.

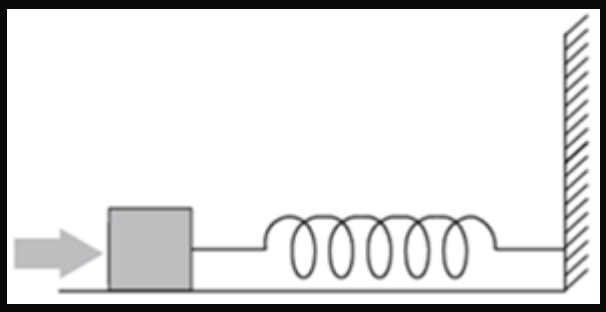
Q.22

A glass hemisphere of refractive index $\frac{4}{3}$ and of radius 4 cm is placed on a plane mirror. A point object placed at distance d on axis of this sphere as shown. If the final image is at infinity then find the value of d in cm.



Q.23

A block of mass $0.18\ \mathrm{kg}$ is attached to a spring of force constant $2\ \mathrm{N\ m^{-1}}$.
The coefficient of friction between the block and the floor is 0.1. Initially the block is at rest and the spring is unstretched. An impulse is given to the block as shown in the figure. The block slides a distance of $0.06\ \mathrm{m}$ and comes to rest for the first time. The initial velocity of the block in $\mathrm{ms^{-1}}$ is $v = N/10$. Then N is



Q.24

A balloon rises from rest with a constant acceleration of $\frac{g}{8}$. A stone is released from it when it has risen to height h. The time taken by the stone to reach the ground is $n\sqrt{\frac{h}{g}}$ seconds. Then what is the value of n ?

Q.25

A thin rectangular magnet suspended freely has a period of oscillation 4 s. If it is broken into two halves each having half their initial length, then when suspended similarly, the time period of oscillation of each part will be

Q.26

The bond order in NO is 2.5 while that in NO^+ is 3. Which statement is true for these two species:

(A) Bond length in NO^+ is greater than that in NO

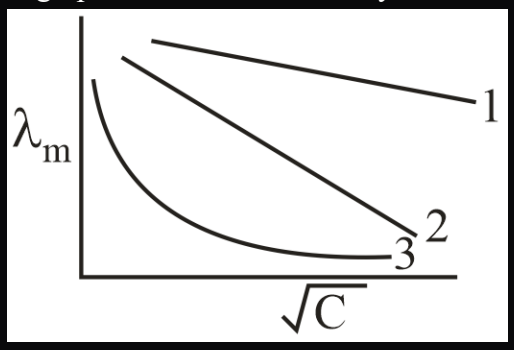
(B) Bond length in NO is greater than that in NO^+

(C) Bond length is NO^+ is equal to that in NO

(D) Bond length is unpredictable

Q.27

A graph of molar conductivity of three electrolytes (NaCl, HCl and $\mathrm{NH_4OH}$) is plotted against $\sqrt{\mathrm{C}}$



Which of the following options is correct?

(A) $\overset{(1)}{\mathrm{NaCl}}\ \overset{(2)}{\mathrm{HCl}}\ \overset{(3)}{\mathrm{NH_4OH}}$

(B) $\overset{(1)}{\mathrm{NH_4OH}}\ \overset{(2)}{\mathrm{NaCl}}\ \overset{(3)}{\mathrm{HCl}}$

(C) $\overset{(1)}{\mathrm{HCl}}\ \overset{(2)}{\mathrm{NaCl}}\ \overset{(3)}{\mathrm{NH_4OH}}$

(D) $\overset{(1)}{\mathrm{NH_4OH}}\ \overset{(2)}{\mathrm{HCl}}\ \overset{(3)}{\mathrm{NaCl}}$

Q.28

Among the following, the reaction that is accompanied by a decrease in the entropy is

(A) $\mathrm{N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)}$

(B) $\mathrm{C_6H_{12}O_6(s) + 6O_2(g) \rightarrow 6CO_2(g) + 6H_2O(l)}$

(C) $\mathrm{PCl_5(s) \rightarrow PCl_3(l) + Cl_2(g)}$

(D) $\mathrm{2H_2O(l) \rightarrow 2H_2(g) + O_2(g)}$

Q.29

$1\ \mathrm{g}$ of a complex $\mathrm{[Cr(H_2O)_5Cl]Cl_2 \cdot H_2O}$ (mol. Wt. 266.5) was passed through a cation exchanger to produce HCl. The acid liberated was diluted to 1 litre. The normality of this acid solution is

(A) $5 \times 10^{-3}\ \mathrm{N}$

(B) $7.5 \times 10^{-3}\ \mathrm{N}$

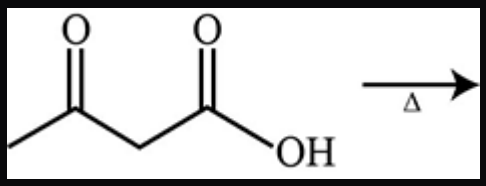
(C) $7.5 \times 10^{-2}\ \mathrm{N}$

(D) $7.5 \times 10^{-1}\ \mathrm{N}$

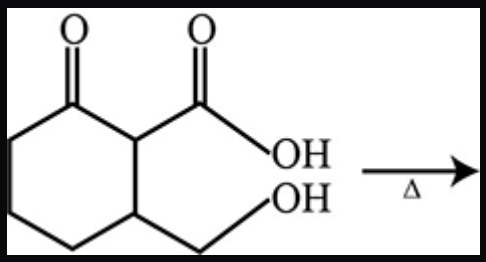
Q.30

In which of the following reaction $\mathrm{CO_2}$ (carbon dioxide) is not released?

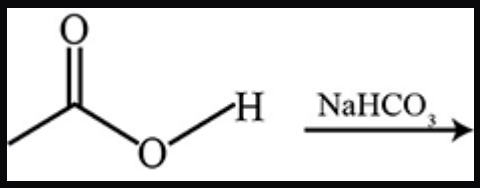
(A)



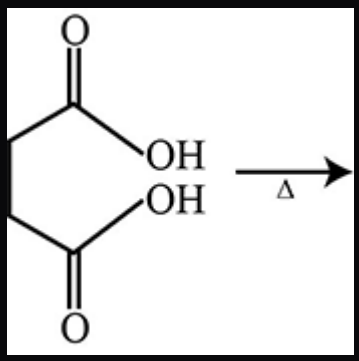
(B)



(C)



(D)



Q.31

The oxidation number of nitrogen atoms in ammonium nitrate are

(A) $-3, +5$

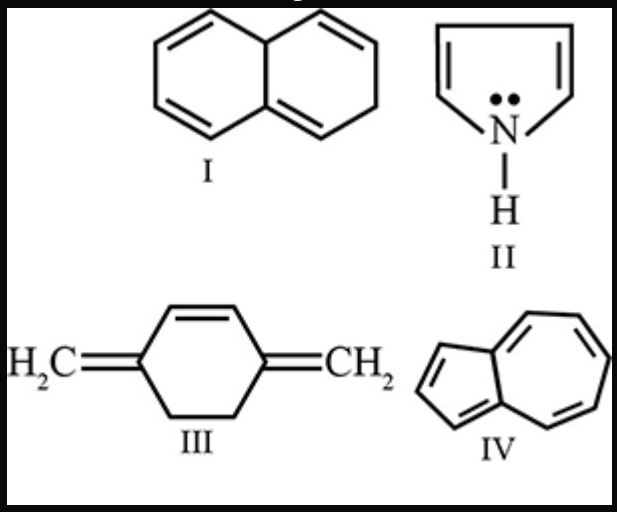
(B) $-3, -3$

(C) $+5, +5$

(D) $+2, -3$

Q.32

Which of the following will exhibit aromatic character?



(A) I, II

(B) III, IV

(C) II, IV

(D) II, III

Q.33

Compounds A and B are treated with dil. HCl separately. The gas liberated are Y and Z respectively. Y turns acidified $K_2Cr_2O_7$ paper green while Z turns lead acetate paper black. The compounds A and B are respectively.

(A) NaCl and Na_2CO_3

(B) Na_2SO_3 and Na_2 S

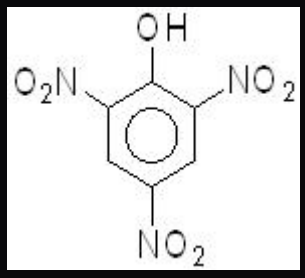
(C) Na_2 S and Na_2SO_3

(D) Na_2SO_3 and Na_2SO_4

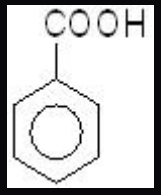
Q.34

Which of the following is not soluble in $NaHCO_3$ solution?

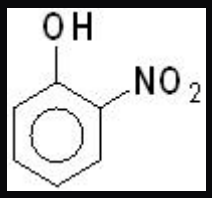
(A)



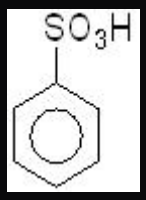
(B)



(C)

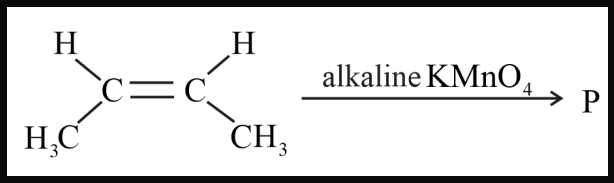


(D)



Q.35

Which of the following statements is true for the reaction given below?



(A) P is a the meso compound 2, 3-butanediol formed by syn-addition

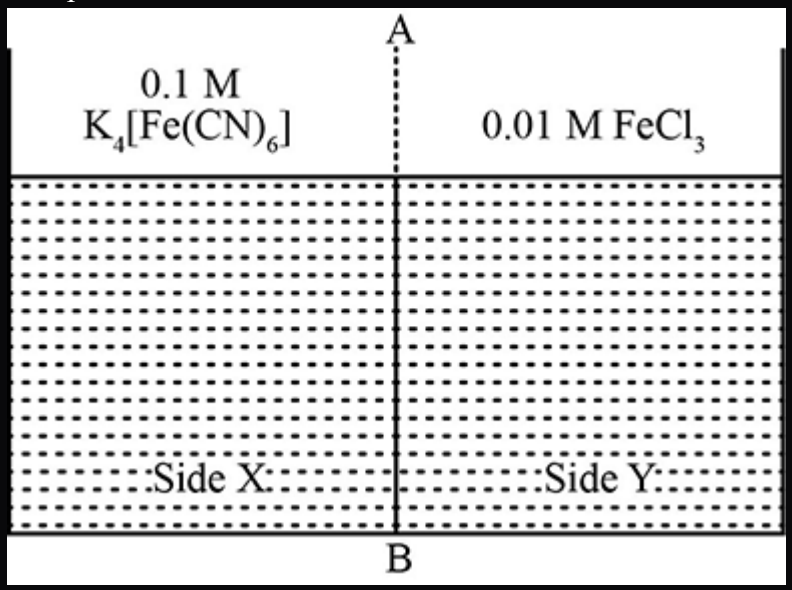
(B) P is a the meso compound 2, 3-butanediol formed by anti-addition

(C) P is a racemic mixture of (d) and (I) 2, 3-butanediol formed by anti-addition

(D) P is a racemic mixture of (d) and (I) 2, 3-butanediol formed by syn-addition

Q.36

$FeCl_3$ on reaction with $K_4\,[Fe(CN)_6]$ in aqueous solution gives blue colour. These are separated by a semipermeable membrane AB as shown. Due to osmosis there is



(A) blue colour formation in side X

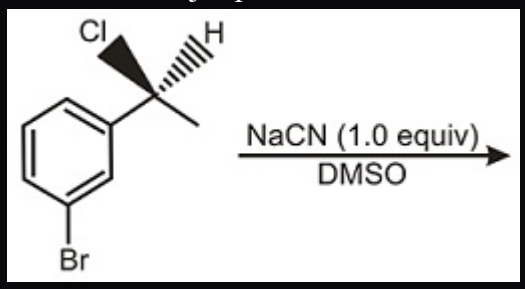
(B) blue colour formation in side Y

(C) blue colour formation in both of the side X and Y

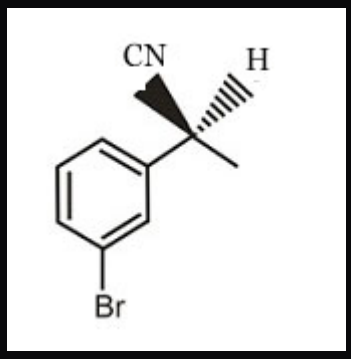
(D) no blue colour formation

Q.37

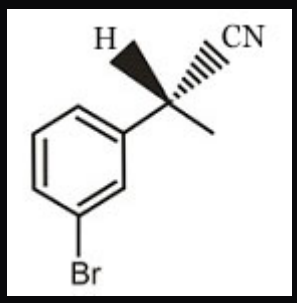
Predict the major product of the following reaction;



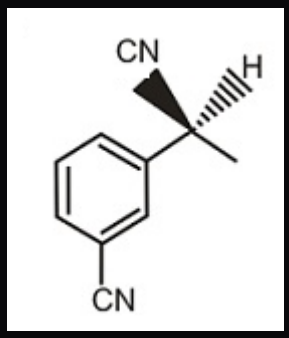
(A)



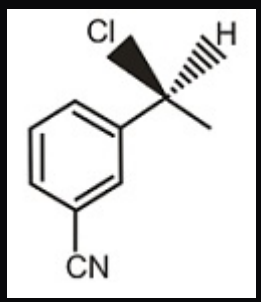
(B)



(C)



(D)



Q.38

The experimental data for the reaction,

$2\ \mathrm{A} + \mathrm{B}_2 \rightarrow 2\mathrm{AB}$ is

Experiment	$[A]$	$[B_2]$	Rate (mole s^{-1})
1	0.50	0.50	1.6×10^{-4}
2	0.50	1.00	3.2×10^{-4}
3	1.00	1.00	3.2×10^{-4}

The rate equation for the above data is

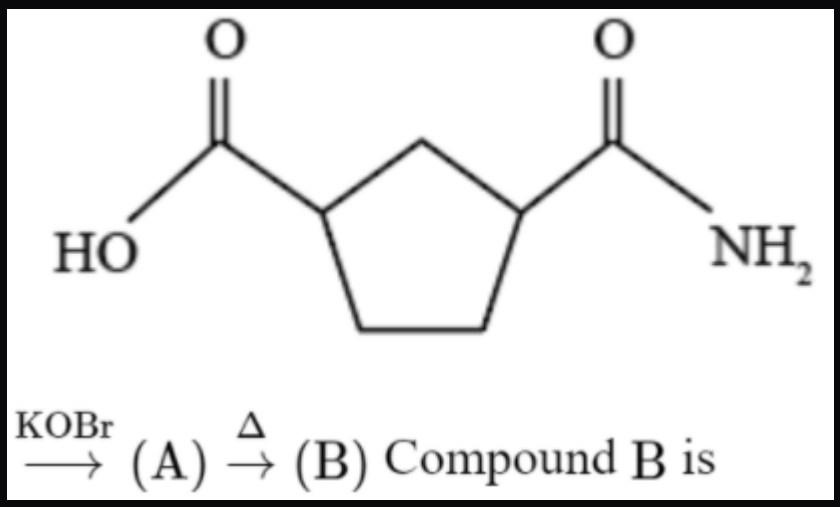
(A) Rate $= k[A]^2[B]^2$

(B) Rate $= k[A]^2[B]$

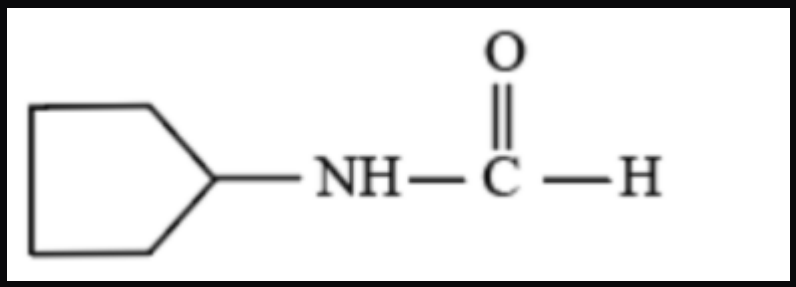
(C) Rate $= k[B_2]$

(D) Rate $= k[B_2]^2$

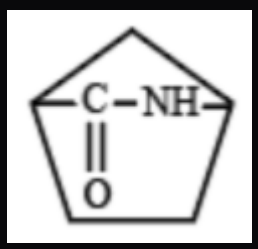
Q.39



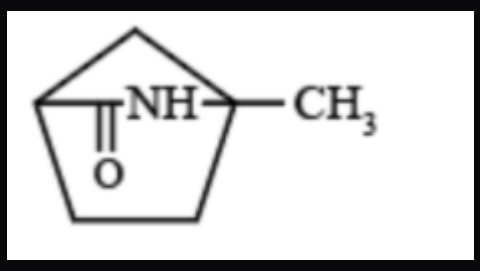
(A)



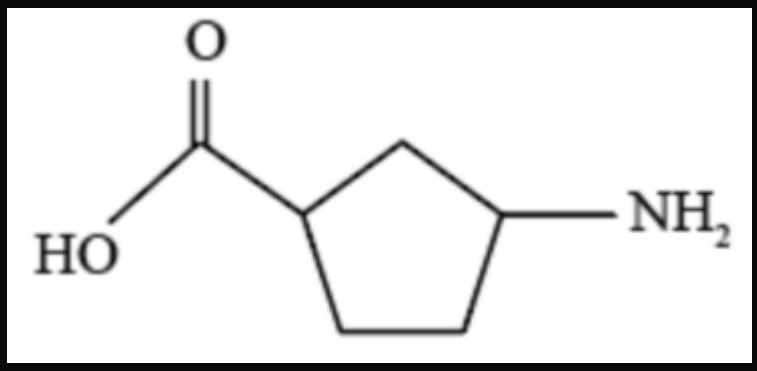
(B)



(C)



(D)



Q.40

Which among the following is an example of ionization isomer?

(A) $[Co(NH_3)_5SO_4]Br$ and $[Co(NH_3)_5Br]SO_4$

(B) $[Co(NH_3)_5NO_2]Cl$ and $[Co(NH_3)_5ONO]Cl_2$

(C) $[Co(H_2O)_6]Cl_3$ and $[Co(H_2O)_5Cl]Cl_2.H_2O$

(D) $[Co(NH_3)_3]$ $[Cr(CN)_6]$ and $[Cr(NH_3)_6]$ $[Co(CN)_6]$

Q.41

For a hypothetical hydrogen like atom, the potential energy of the system is given by $U(r) = \frac{-ke^2}{r^4}$, where r is the distance between the two particles. If Bohr's model of quantisation of angular momentum is applicable, then velocity of particle is given by

(A) $\frac{nh}{16ke\pi^2m^{3/2}}$

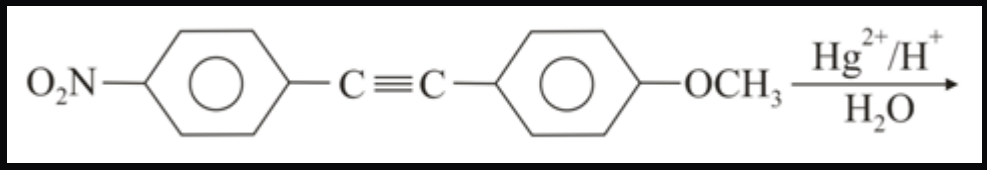
(B) $\frac{n^2h^2}{8k^2e^2\pi^4m^3}$

(C) $\frac{n^3h^3}{2k^2e^3\pi^3m^4}$

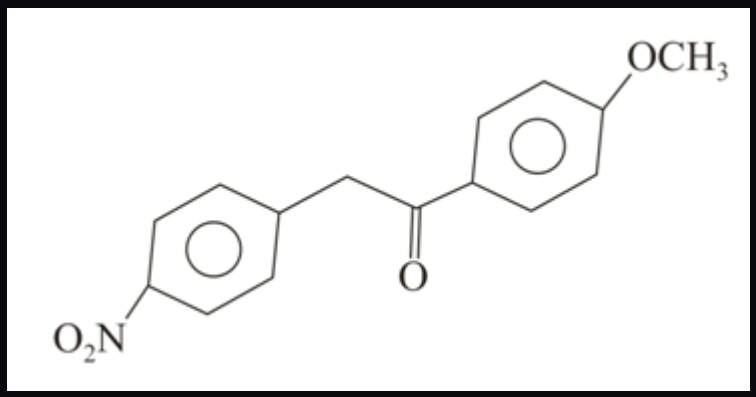
(D) $\frac{n^2h^2}{8k^{1/2}e\pi^2m^{3/2}}$

Q.42

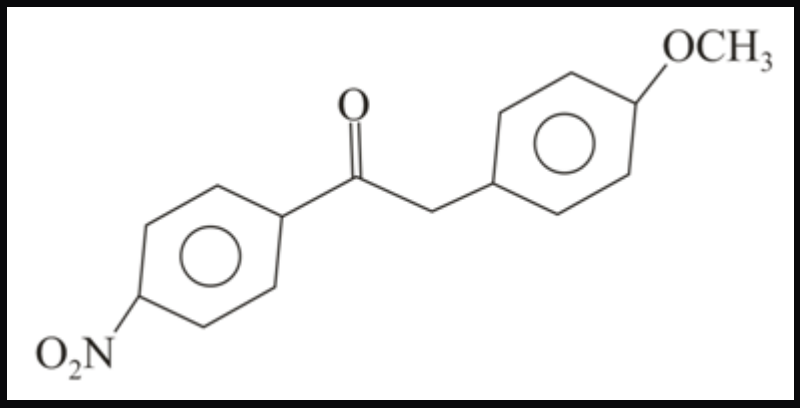
The major product obtained from the following reaction is :



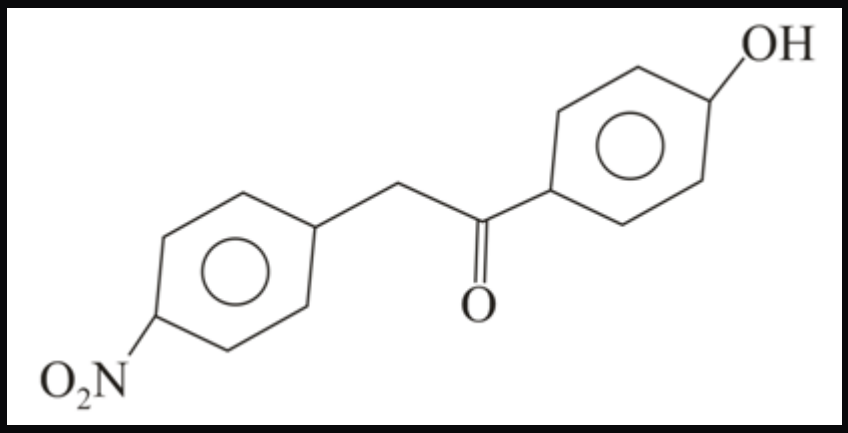
(A)



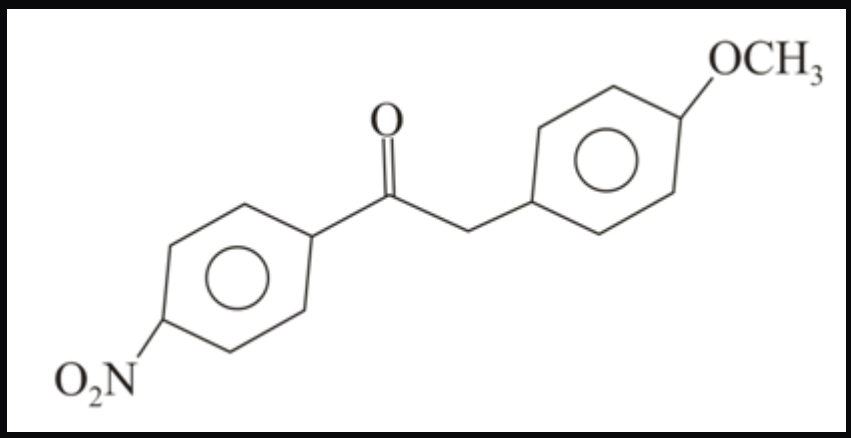
(B)



(C)

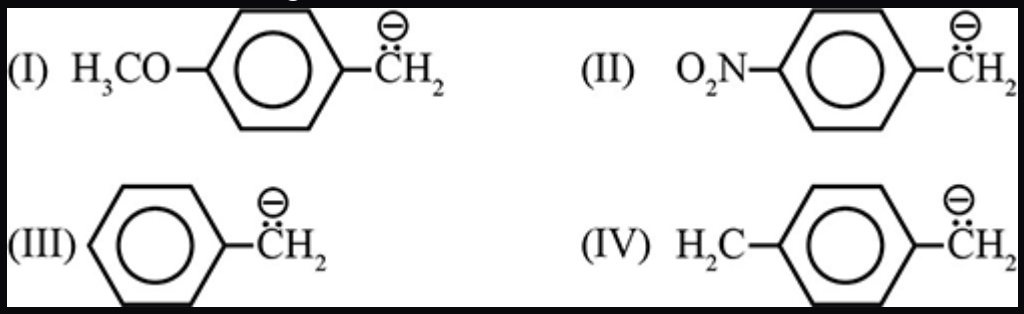


(D)



Q.43

Consider the following carbanions:



Correct decreasing order of stability is -

(A) $II > III > IV > I$

(B) $III > IV > I > II$

(C) $IV > I > II > III$

(D) $I > II > III > IV$

Q.44

Match the molecules given from (List I) with the shape obtained through VSEPR theory (List II).

	List (I) Molecules		List (II) Shapes
A	PCl_5	i	V-Shaped
B	OF_2	ii	Triangular planar
C	BCl_3	iii	Trigonal bipyramidal
D	NH_3	iv	Trigonal pyramidal
		v	Tetrahedral

(A) $\mathrm{A-i,\ B-v,\ C-iv,\ D-iii}$

(B) $\mathrm{A-ii,\ B-iii,\ C-i,\ D-ii}$

(C) $\mathrm{A-iv,\ B-iii,\ C-ii,\ D-v}$

(D) $\mathrm{A-iii,\ B-i,\ C-ii,\ D-iv}$

Q.45

Which of the following is a wrong statement about Oxymercuration-Demercuration?

(A) In the 1st step oxymercuration occurs i.e., water and $Hg\ (OAc)_2$ add to double bond

(B) In the 2nd step demercuration occurs i.e., $NaBH_4$ reduces $HgOAc$group to hydrogen

(C) The net reaction is addition of water according to Markovnikov's rule

(D) Rearrangement takes place

Q.46

The following reaction occurs during rusting of iron

$2H^+ + 2e + \frac{1}{2}O_2 \rightarrow H_2O; E^o = +1.23\ V$

$Fe^{2+} + 2e \rightarrow Fe(s); E^0 = -0.44\ V$

Calculate magnitude of $\Delta G^\circ(kJ)$ for the net process

$Fe(s) + 2H^+ + \frac{1}{2}O_2 \rightarrow Fe^{2+} + H_2O$

(Take $1\ F = 96500$ and mark answer to nearest whole number)

Q.47

A gaseous mixture of three gases A, B and C has a pressure of $10\ atm$. The total number of moles of all the gases is 10 . If the partial pressures of A and B are 3.0 and 1.0 atm respectively and if ' C ' has mol. wt of 2.0 what is the weight of ' C ' in g present in the mixture.

Q.48

How many total number of elements (upto atomic number = 10) are here, which have positive electron gain enthalpy (EGE_1)?

Q.49

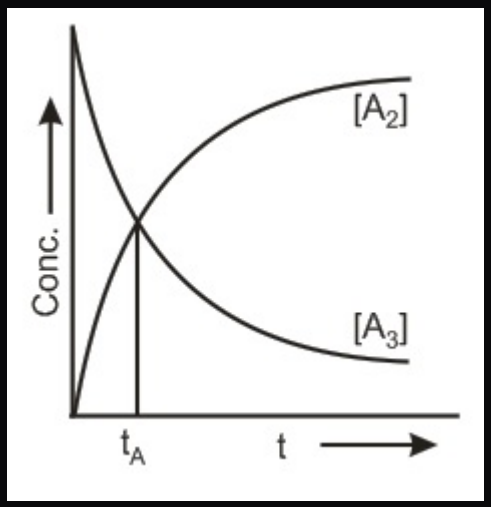
How many isomers of $C_5H_{11}Br$ show stereochemical inversion when it undergoes SN^2 reaction?

Q.50

Consider a first order decomposition process :

$A_3 \rightarrow \frac{3}{2}A_2$

A plot of concentration of A_3 and A_2 versus time is shown below. At time t_A, percentage of reactant decomposed is ;:



Q.51

$\lim_{n\to\infty} \prod_{r=3}^{n} \frac{(r+2)\left(r^2-2r+2^2\right)}{(r-2)(r^2+2r+2^2)}$

(A) $\frac{3}{2}$

(B) $\frac{5}{2}$

(C) $\frac{7}{2}$

(D) $\frac{9}{2}$

Q.52

The point on the line $\frac{x-2}{1}=\frac{y+3}{-2}=\frac{z+5}{-2}$ at a distance of 6 from the point $(2,-3,-5)$ is

(A) $(3,-5,-3)$

(B) $(4,-7,-9)$

(C) $(0,2,-1)$

(D) $(-3,5,3)$

Q.53

If $f:[1,\infty) \rightarrow R: f(x)$ is monotonic and differentiable function and $f(1)=1$, then number of solutions of the equation $f(f(x))=\frac{1}{x^2-2x+2}$ is/are ....

(A) 2

(B) 0

(C) 1

(D) 4

Q.54

The solution of differential equation $2\left(x-y\frac{dy}{dx}\right)\left(x^2+y^2\right)=\left(x^2-y^2\right)\left(y-x\frac{dy}{dx}\right)$ is

(A) $\log\left|x^2-y^2\right|+\tan^{-1}\frac{y}{x}=C$

(B) $\log\left|x^2-y^2\right|-\tan^{-1}\frac{y}{x}=C$

(C) $\log\left|\frac{y}{x}\right|-\tan^{-1}\left(x^2-y^2\right)=C$

(D) $\log\left|\frac{y}{x}\right|+\tan^{-1}\left(x^2-y^2\right)=C$

Q.55

A man moves one unit distance for each step he takes either along the positive x-direction or negative x-direction. The number of ways he moves from the point $(0,0)$ to $(2,0)$ in at most 10 steps is

(A) 212

(B) 225

(C) 256

(D) 286

Q.56

The distance between circumcentre and orthocentre of the triangle whose vertices are $O(0,0), A(6,8)$ and $B(-4,3)$ is

(A) $\frac{5\sqrt{5}}{4}$

(B) $\frac{5\sqrt{5}}{12}$

(C) $\frac{5\sqrt{5}}{2}$

(D) $5\sqrt{5}$

Q.57

If $f(x)=\int 2^{\frac{\sin x}{\log_e 2}}\left(\frac{5\cos x+\cos^3 x}{(2-\sin x)\sqrt{3+\cos^2 x}}\right)dx$ and $f(0)=1$ then value of $f(\pi/2)$ equals

(A) $\sqrt{3}\mathrm{e}$

(B) $\frac{\sqrt{3}\mathrm{e}}{2}$

(C) $\frac{3\mathrm{e}}{2}$

(D) $3\mathrm{e}$

Q.58

If a and c are unit vectors and $|\vec{b}|=\sqrt{5}$. The angle between $\vec{a}$ and $\vec{c}$ is $\frac{\pi}{3}$. and $\vec{b}-2\vec{c}=\alpha\vec{a}$ Then, α is a root of the equation:

(A) $x^2+2x+1=0$

(B) $x^2+x+1=0$

(C) $x^2+x+12=0$

(D) $x^2+2x-1=0$

Q.59

Let A be a 3×3 matrix such that $a_{11}=a_{33}=2$ and all the other $a_{ij}=1$. Let $A^{-1}=x\,A^2+yA+zI$ then find the value of $(x+y+z)$ where I is a unit matrix of order 3.

(A) 2

(B) 5

(C) 1

(D) 3

Q.60

If A_1, A_2, A_3, A_4 are any four consecutive binomial coefficients in the expansion of $(1+x)^n$, then $\frac{A_1}{A_1+A_2}+\frac{A_3}{A_3+A_4}-\frac{2A_2}{A_2+A_3}=$

(A) $n-2$

(B) 1

(C) -1

(D) 0

Q.61

Let $\alpha, \beta \in \mathrm{R}$, satisfy the equations $\cot^2 \alpha - \operatorname{cosec}^2 \beta = \frac{2k}{3} - 5$ & $\cot^2 \beta - \operatorname{cosec}^2 \alpha = \frac{k^2}{2}$, then product of all possible value(s) of k is equal to

(A) $-\frac{4}{3}$

(B) -6

(C) -1

(D) $\frac{2}{3}$

Q.62

Let n be a fixed positive integer. Define a relation R in the set Z of integers by aRb if and only if $a - b$ is divided by n. The relation R is

(A) reflexive

(B) symmetric

(C) transitive

(D) an equivalence relation

Q.63

If $\alpha, \beta, \gamma, \delta$ are roots of equation $\mathrm{x}^4 - 7\mathrm{x}^2 + \mathrm{x} - 5 = 0$, then the value of $(\alpha + \beta + \gamma)(\alpha + \beta + \delta)(\beta + \gamma + \delta)(\alpha + \gamma + \delta)$ is-

(A) -1

(B) 1

(C) 5

(D) -5

Q.64

Let $A \equiv (-4, 0)$ and $B \equiv (4, 0)$ if the number of points C on the circle $x^2 + y^2 = 16$, such that the area of triangle whose vertices are A, B and C is positive integer is N then value of $\left[\frac{\mathrm{N}}{8}\right]$ is equal to ? (where [] denotes the greatest integer function)

(A) 4

(B) 5

(C) 3

(D) 7

Q.65

Let A and B be two events such that $P(A \cup B) \geq 3/4$ and $1/8 \leq P(A \cap B) \leq 3/8$
Statement $-1:\quad P(A)+P(B) \geq 7/8$
Statement $-2:\quad P(A)+P(B) \leq 11/8$

(A) Only I is true

(B) Both I, II are true

(C) Only II is true

(D) neither I nor II is true

Q.66

If 1 lies between the roots of the equation $y^2 - my + 1 = 0$ and [.] denotes the greatest integer function, then $\int_0^{10} \left[\left(\frac{4|x|}{x^2+16}\right)^m\right] dx$ is equal to

(A) 0

(B) 2

(C) 10

(D) None of these

Q.67

Let $\mathrm{a}, \mathrm{b}, \mathrm{c} \in R^+$ such that $\mathrm{a}+\mathrm{b}+\mathrm{c}=6$ then the range of ab^2c^3 is

(A) $(0, \infty)$

(B) $(0, 1)$

(C) $(0, 108]$

(D) $(1, 96]$

Q.68

If z is a complex number such that $\mathrm{z}+\mathrm{i}|\mathrm{z}| = \mathrm{i}\bar{\mathrm{z}}+1$, then $|\mathrm{z}|$ is -

(A) 4

(B) 2

(C) 1

(D) 0

Q.69

Equation of the line of shortest distance between the lines $\frac{x}{2} = \frac{y}{-3} = \frac{z}{1}$ and $\frac{x-2}{3} = \frac{y-1}{-5} = \frac{z+2}{2}$ is -

(A) $3(x-21) = (3y-92) = (3z-32)$

(B) $3x-62=3y-93=3z+31$

(C) $\frac{(x-21)}{3}=\frac{\left(y+\frac{92}{3}\right)}{3}=\frac{\left(z-\frac{32}{3}\right)}{3}$

(D) $x-\frac{62}{3}=y+31=\left(z+\frac{31}{3}\right)$

Q.70

If $f(x+y)=f(x)+f(y)-xy-1, \forall x, y \in R$ and $f(1)=1$, then the number of solutions of $f(n)=n, n \in N$, is

(A) one

(B) two

(C) no solution

(D) three

Q.71

Let one observation equal to the mean is added to n observations. If the variance changes from 78 to 72, then n is equal to

Q.72

Let P, Q be points of the ellipse $16x^2+25y^2=400$, so that $PQ=96/25$ and P, Q lie above major axis. The circle drawn with PQ as diameter touches major axis at positive focus. If m is slope of PQ then the value of $1/|m|$ is ____.

Q.73

$A=\begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$ where $\mathrm{a}, \mathrm{b}, \mathrm{c}$ are real positive numbers such that $abc=1$ and $AA^T=I$ then $a^3+b^3+c^3=$

Q.74

If $\int_{\frac{\pi}{4}}^{\frac{\pi}{3}} \frac{\left(\sin^3\theta-\cos^3\theta-\cos^2\theta\right)\left(\sin\theta+\cos\theta+\cos^2\theta\right)^{2007}}{(\sin\theta)^{2009}(\cos\theta)^{2009}}d\theta=\frac{(a+\sqrt{b})^d-(1+\sqrt{c})^d}{d}$ where a, b, c, d are all positive integers then $a+b+c+d=$ ____

Q.75

If area bounded by curve $y=\left|\cos^{-1}(\sin x)\right|+\left|\sin^{-1}(\cos x)\right|$, x-axis and $\frac{\pi}{2} \leq x \leq \pi$ is equal to $\frac{\pi^2}{\mathrm{k}}$ (where $k \in I$), then k is

Answers & Solutions

Q.1 Answer:

$\frac{hc}{3\lambda}$

Solution:

Let W be the work-function which corresponds to the energy required to pick out an electron from a materia then, $\Rightarrow \quad \mathrm{KE} = E = \frac{hc}{\lambda} - W$

$\Rightarrow 4E = \frac{3hc}{\lambda} - W$

$\Rightarrow \frac{1}{4} = \frac{\frac{hc}{\lambda} - W}{\frac{3hc}{\lambda} - W}$ [from (i) and (ii)]

$\Rightarrow W = \frac{hc}{3\lambda}$

Q.2 Answer:

Assertion is true but reason is false.

Solution:

$f = \frac{v}{4(l+0.6r)}$ or $\frac{v}{2(l+1.2r)}$

Therefore, if r is less, f will be more.

Q.3 Answer:

$\frac{105}{11}$ A

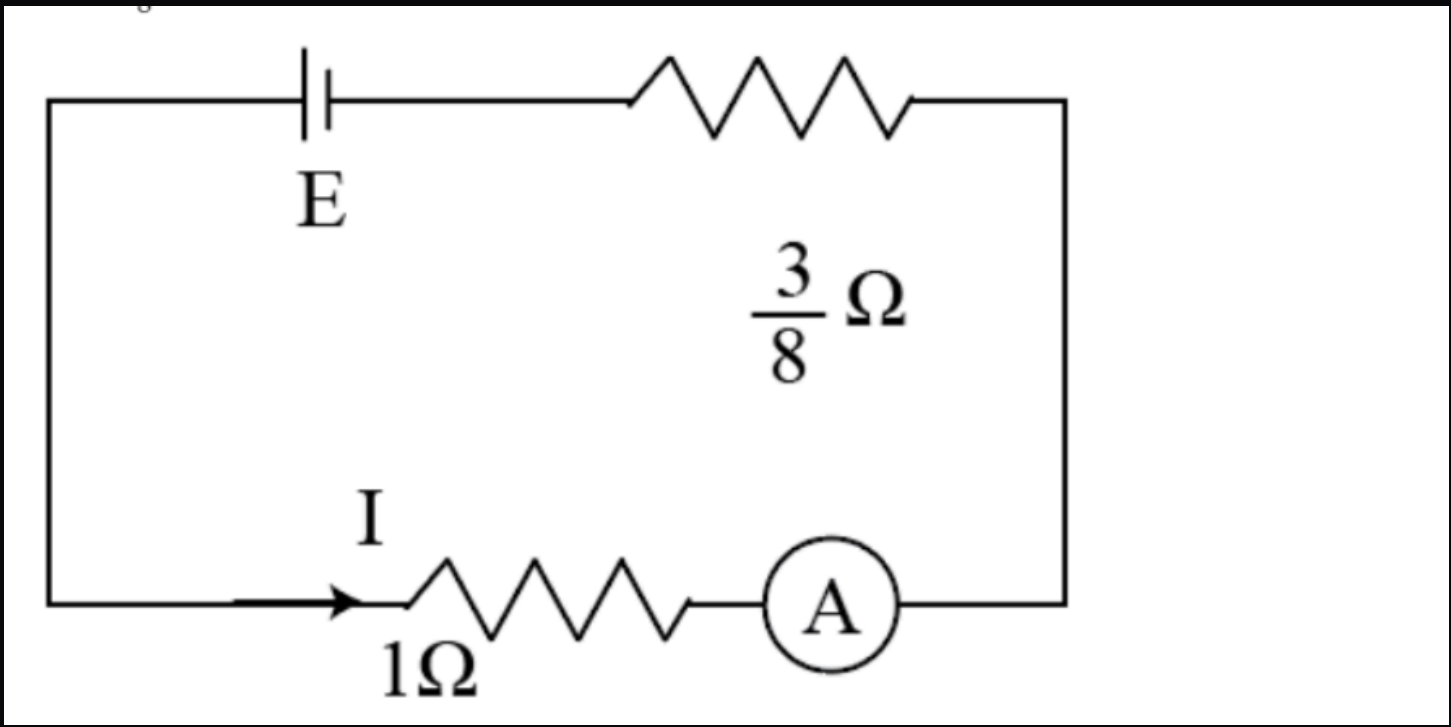
Solution:

For cells in parallel $\frac{1}{r} = \frac{1}{r_1} + \frac{1}{r_2} = \frac{1}{1} + \frac{1}{0.6}$ $\frac{1}{r} = 1 + \frac{5}{3} = \frac{8}{3}$ $r = \frac{3}{8}\Omega$ $\frac{E}{r} = \frac{E_1}{r_1} + \frac{E_2}{r_2} \Rightarrow \frac{8E}{3} = \frac{10}{1} + \frac{15}{0.6} \Rightarrow \frac{8E}{3} = 35 \Rightarrow E = \frac{105}{8}V$ $I = \frac{E}{\frac{3}{8}+1} = \frac{105}{11}A$

$r = \frac{3}{8}\Omega$

$\frac{E}{r} = \frac{E_1}{r_1} + \frac{E_2}{r_2} \Rightarrow \frac{8E}{3} = \frac{10}{1} + \frac{15}{0.6} \quad \Rightarrow \quad \frac{8E}{3} = 35 \Rightarrow E = \frac{105}{8}V$

$I = \frac{E}{\frac{3}{8}+1} = \frac{105}{11}A$



Q.4 Answer:

$\frac{A\alpha\left(T_1^2-T_2^2\right)}{2l}$

Solution:

Heat current,

$$i=-KA\frac{dT}{dX}$$

$$\mathrm{idx}=-\mathrm{KAdT}$$

$$i\int_0^l \mathrm{dX}=-\mathrm{A}\alpha\int_{\mathrm{T}_1}^{\mathrm{T}_2}\mathrm{TdT}$$

$$\Rightarrow \mathrm{i}l=-\mathrm{A}\alpha\frac{\left(\mathrm{T}_2^2-\mathrm{T}_1^2\right)}{2}$$

$$\Rightarrow \mathrm{i}=\mathrm{A}\alpha\frac{\left(\mathrm{T}_1^2-\mathrm{T}_2^2\right)}{2l}$$

Q.5 Answer:

Velocity of light for violet is less than the velocity of light for red.

Solution:

As $\mu_{\mathrm{v}}>\mu_{\mathrm{r}}$ **therefore,** $\mathrm{v_V}<\mathrm{v_r}$

Q.6 Answer:

2.87 and 2.83

Solution:

For caliper C_1

$10\ \mathrm{VSD}=9\ \mathrm{MSD}$

$\mathrm{LC}=1\ \mathrm{MSD}-1\ \mathrm{VSD}$

$\mathrm{LC}=0.01\ \mathrm{cm}$

Measured value=Main scale reading+vernier scale reading×least count

$=(2.8+7\times 0.01)\ \mathrm{cm}$

$=2.87\ \mathrm{cm}$

For caliper C_2

$10\ \mathrm{VSD}=11\ \mathrm{MSD}$

$\mathrm{LC}=-0.01\ \mathrm{cm}$ **(Negative least count means we have to take VSR from the right of the scale**

Measured value $=\{2.8+(10-7)\times 0.01\}\ \mathrm{cm}$

$=2.83\ \mathrm{cm}$

Q.7 Answer:

$\left[\mathrm{L}^{-2}\mathrm{TA}\right]$

Solution:

Electric displacement, $D = \varepsilon E$

Unit of $D = \frac{\mathrm{C}^2}{\mathrm{Nm}^2}\frac{\mathrm{N}}{\mathrm{C}}$

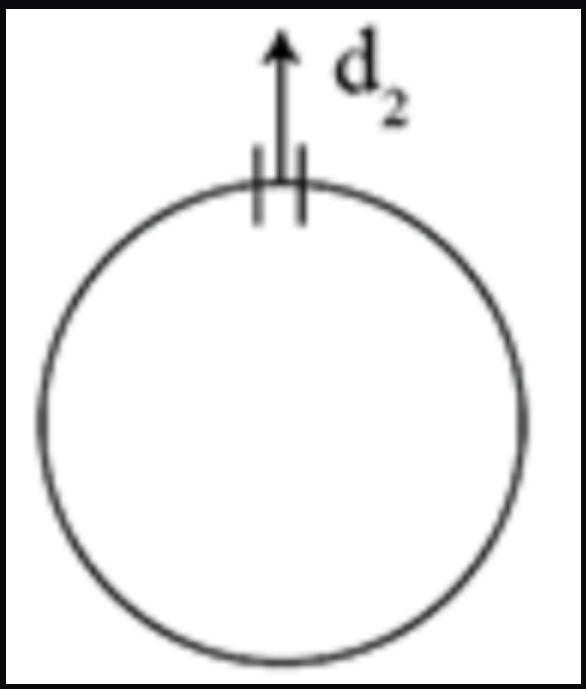
$\therefore \quad [D] = \left(\frac{\mathrm{C}}{\mathrm{m}^2}\right) = \frac{[\mathrm{AT}]}{[\mathrm{L}^2]} = \left[\mathrm{L}^{-2}\mathrm{TA}\right]$

Q.8 Answer:

$0.5N/m^2$

Solution:

$$dF = dq \times \frac{\sigma}{2\varepsilon_0}$$
$$E = \frac{\sigma}{2\varepsilon_0}$$
$$dF = \sigma dA \times \frac{\sigma}{2\varepsilon_0}$$
$$\Rightarrow \frac{dF}{dA} = \frac{\sigma^2}{2\varepsilon_0}$$
$$P = \frac{\sigma^2}{2\varepsilon_0}$$



Q.9 Answer:

$3 \times 10^{10}\ \mathrm{N\ m^{-2}}$

Solution:

$\mathrm{Stress} = Y \times \mathrm{strain} = 2 \times 10^{11} \times 0.15\ \mathrm{N\ m^{-2}}$
$= 3 \times 10^{10}\ \mathrm{N\ m^{-2}}.$

Q.10 Answer:

1, 2, 4

Solution:

As the total number of wavelengths emitted is 6, let the final excited state of the be n, then

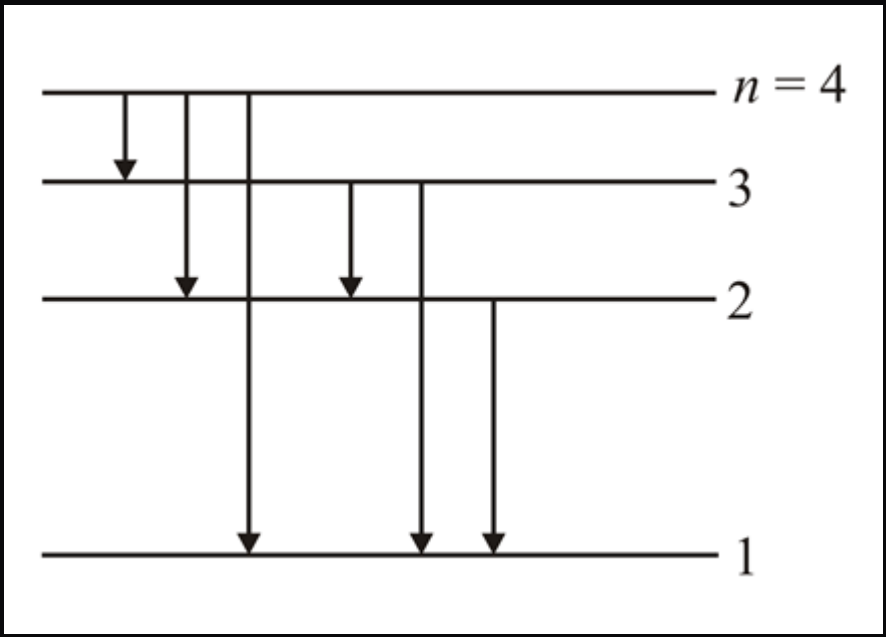
$\frac{n(n-1)}{2} = 6 \Rightarrow n = 4$

Hence, the final excited state for the atom is $n = 4$

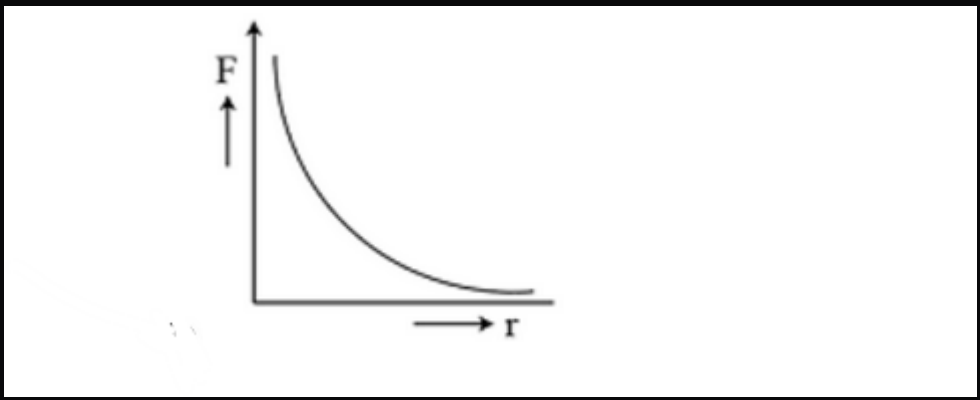
If the initial state were $n = 3$, in the emission spectrum, then no wavelengths shorter than λ_0 would have occurred. But if we take the initial state to be $n = 2$, then the transition from $n = 2$ to $n = 3$ and from $n = 3$ to $n = 4$ have wavelengths shorter than λ_0.

Lymann series is the spectral series that occurs when transition takes place from $n \geq 2$ to $n = 1$, therefore in this case, we have 3 spectral lines in Lymann series.

$\therefore$ (A), (B) and (D)



Q.11 Answer:



Solution:

$F = ma = \frac{mv^2}{r} \Rightarrow F \propto \frac{1}{r} \Rightarrow$ Rectangular hyperbola.

Q.12 Answer:

$\frac{\pi}{2}$

Solution:

Let the angular speed of the system just after the collision be ω.

Applying conservation of angular momentum about the centre of the rod, we get

$2mvr_{\perp} = I\omega$

$\Rightarrow 2 \times \frac{M}{6} \times L \times \frac{L}{2} = \left[\frac{ML^2}{12} + \frac{M}{6}\left(\frac{L}{2}\right)^2 \times 2\right]\omega$

$\Rightarrow \frac{ML^2}{6} = \frac{ML^2}{6}\omega$

$\Rightarrow \omega = 1 \text{ rad s}^{-1}$

Time to rotate $\theta = \frac{\pi}{2}$

$t = \frac{\theta}{\omega} = \frac{\pi/2}{1} = \frac{\pi}{2}\ s.$

Q.13 Answer:

$2\pi \times 10^{-3}$ T

Solution:

Given that length of solenoid $(l) = 1.5$ m

number of turns$(n) = 10\ ^{\text{turns}}/_{\text{cm}} = 1000\ ^{\text{turns}}/_{\text{m}}$

current $(i) = 5$ A, **diameter** $(d) = 1.5$ m

The magnetic field induction at axis inside the solenoid is$(B), B = \mu_0 ni = 4\pi \times 10^{-7} \times (10 \times 100) \times 5$

$\Rightarrow B = 2\pi \times 10^{-3}$ T

Q.14 Answer:

$\frac{\omega^2 R^2}{2g}$

Solution:

The value of acceleration due to gravity at the equator is $g_E = g - R\omega^2$ **and the acceleration due to gravity at a height** h **above the pole is** $g_h = \left(1 - \frac{2h}{R}\right) g$**. It is given that**

$\because mg_E = mg_h$

$\therefore g_E = g_h$

$\therefore g_E = g_h$

$$\therefore g - R\omega^2 = \left(1 - \frac{2h}{R}\right) g = g - \frac{2hg}{R}$$

$$\therefore R\omega^2 = \frac{2hg}{R} \therefore h = \frac{R^2\omega^2}{2g}$$

Q.15 Answer:

$A \to s, B \to p, C \to r, D \to q$

Solution:

Diode with no biasing $\Rightarrow \text{I}_{\text{drift}} = \text{i}_{\text{diffusion}}$

Diode in forward bias $\Rightarrow i_{diffusion} >> i_{drift}$

Diode in reverse bias without breakdown

$\Rightarrow i_{drift} > i_{diffusion}$

Diode in reverse bias with breakdown

$\Rightarrow i_{drift} >> i_{diffusion}$

Q.16 Answer:

$\frac{4}{3}\mu F$

Solution:

Starting from far end :

$1\mu F$ **and** $2\mu F$ **in parallel** $\equiv 1+2=3\mu F$

Now, 3 capacitors of $3\mu F$ **in series** $\equiv \frac{3}{3}=1\mu F$

Now, $l\mu F, l\mu F$ **and** $2\mu F$ **in parallel** $\equiv 1+1+2=4\mu F$

Now, 3 capacitors of $4\mu F$ **each in series** $\equiv \frac{4}{3}\mu F$

Q.17 Answer:

$10\ \text{cm}, \frac{2}{3}\ \text{s}$

Solution:

The equation for SHM is given as

$y=5\left(\sin 3\pi t+\sqrt{3}\cos 3\pi t\right)$

$=5\times 2\left(\frac{1}{2}\times \sin 3\pi t+\frac{\sqrt{3}}{2}\cos 3\pi t\right)$

$=5\times 2\left(\cos\frac{\pi}{3}\sin 3\pi t+\sin\frac{\pi}{3}\cos 3\pi t\right)$

$=5\times 2\sin\left(3\pi t+\frac{\pi}{3}\right)$

[using, $\sin(a+b)=\sin a\cos b+\cos a\sin b$**]**

Thus, $y=10\sin\left(3\pi t+\frac{\pi}{3}\right)\quad \ldots(i)$

Comparing this equation (i) **with the general equation of SHM, i.e.**$y=A\sin(\omega t+\phi)$

We get, amplitude, $A=10\ \text{cm}$ **and** $\omega=3\pi$**.**

We know that angular frequency and time period are related as $T=\frac{2\pi}{\omega}$**.**

Thus, the time period, $T=\frac{2\pi}{3\pi}=\frac{2}{3}\ \text{s}$**.**

Q.18 Answer:

9

Solution:

As the amplitudes are A **and** $2A$ **then the ratio of intensities is** $1:4$

$I_{\max}=I_0=I_1+I_2+2\sqrt{I_1\times I_2}=I+4I+2\times 2I$

$I_{\max}=I_0=I_1+I_2+2\sqrt{I_1}$

$I_0=9I\Rightarrow I=\dfrac{I_0}{9}$

Intensity at any point:

$I'=I_1+I_2+2\sqrt{I_1\times I_2}\cos\phi$

$I'=I+4I+2\sqrt{I\times 4I}\cos\phi$

$I'=5I+4I\cos\phi; I'=I[5+4\cos\phi]$

$I'=\dfrac{I_0}{9}[5+4\cos\phi]=\dfrac{I_0}{P}[5+4\cos\phi]$

$P=9$

Q.19 Answer:

$\mathrm{P_A = P_B, V_A \neq V_B}$

Solution:

According to problem mass of gases are equal so number of moles will not be equal i.e. $\mu_A \neq \mu_B$
From ideal gas equation $PV = \mu RT \Rightarrow \frac{P_A V_A}{\mu_A} = \frac{P_B V_B}{\mu_B}$
[As temperature of the container are equal]
From the relation it is clear that if $P_A = P_B$ **then**
$\frac{V_A}{V_B} = \frac{\mu_A}{\mu_B} \neq 1$ **i.e.** $V_A \neq V_B$
Similarly if $V_A = V_B$ **then** $\frac{P_A}{P_B} = \frac{\mu_A}{\mu_B} \neq 1$ **i.e.** $P_A \neq P_B$

Q.20 Answer:

$f_r < f_v < f_{uv}$

Solution:

$\lambda_{uv} < \lambda_v < \lambda_r$

$f_{uv} > f_v > f_r$

Q.21 Answer:

—

Solution:

$* Q = CE \Rightarrow 10^{-9} = 10^{-11}(E) \Rightarrow E = 100 \text{ volts}$

$* \omega = \dfrac{1}{\sqrt{LC}} = 10^8 \text{rad|sec}$

$I_{\text{Max}} = \dfrac{E_0}{R} = 1 \text{ A}$

$V_{\text{Max}}^C = I_{\text{Max}} \dfrac{1}{\omega \text{C}} = 1 \times 10^{-8} \times 10^{11} = 1000 \text{ volts}$

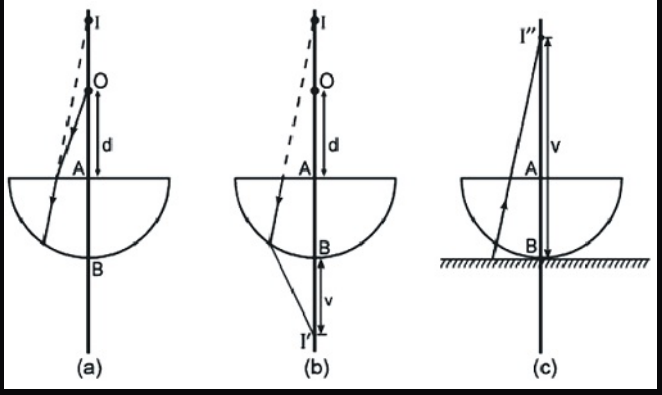
$q_{\text{Max}_C} = CV = 10^{-11} \times 10^3 = 10^{-8} \text{C}$

Q.22 Answer:

—

Solution:

The image of object O by refraction at plane surface is formed at I such that
$Ai = \frac{4}{3} d$



I acts as object for curved surface. The curved surface makes image o I at I'

$\frac{1}{v}-\frac{\frac{4}{3}}{-\left(R+\frac{4}{3}d\right)}=\frac{1-\frac{4}{3}}{-R}$ or $\frac{1}{v}=\frac{1}{3R}-\frac{4}{3R+4d}$

I' acts as object for mirror. Mirror makes its image at I'' distance v above B

I'' acts as virtual object for the curved surface which makes its image at infinity

$\frac{\frac{4}{3}}{\infty}-\left[\frac{1}{3R}-\frac{4}{3R-4d}\right]=\frac{\frac{1}{3}}{R}$

Solving we get $d=\frac{3}{4}R=\frac{3}{4}\times 4=3cm$

Q.23 Answer:

—

Solution:

From the work-energy principle for the given system,

$$-\frac{1}{2}kx^2-\mu mgx=0-\frac{1}{2}mv^2$$

$$\Rightarrow \quad v=\frac{4}{10}, \quad \text{so } N=4$$

Q.24 Answer:

—

Solution:

From equation $v^2=u^2+2as$

Given, $u=0, a=\frac{g}{8}$

$\therefore$ We have, $v=\sqrt{2\left(\frac{g}{8}\right)h}$

When the stone released from this balloon. It will go upward with velocity $v=\sqrt{gh}$

In this condition, time taken by stone to reach the ground,

$$-h = vt - \frac{1}{2}gt^2$$

$$t = \frac{v}{g}\left[1 + \sqrt{1 + \frac{2gh}{v^2}}\right]$$

$$= \frac{\frac{\sqrt{gh}}{g}}{g}\left[1 + \sqrt{1 + \frac{2gh}{\frac{gh}{4}}}\right]$$

$$= \frac{2\sqrt{gh}}{g} = 2\sqrt{\frac{h}{g}}$$

Q.25 Answer:

—

Solution:

The time period of the dipole in a magnetic field B is given by,

$T = 2\pi\sqrt{\frac{I}{MB}}$; where I is the moment of inertia of bar magnet and M is the effective magnetic moment of magnet.

$T \propto \sqrt{\frac{I}{M}} \cdots (\mathrm{i})$

Given: $T_1 = 4\ \mathrm{s}, I_1 = I$ and $M_1 = M$.

When the magnet is broken into two parts, the moment of inertia becomes $I_2 = \frac{I}{8}$, because the mass is halved and the length also becomes halved. The magnetic moment will become $M_2 = \frac{M}{2}$; so from equation (i),

$$\frac{T_1}{T_2} = \sqrt{\frac{T_1}{M_1}} \times \sqrt{\frac{M_2}{I_2}} = \sqrt{\left(\frac{I \times \frac{M}{2}}{M \times \frac{I}{8}}\right)}$$

$$\frac{4}{T_2} = 2$$

$$T_2 = 2\ \mathrm{s}.$$

Q.26 Answer:

Bond length in NO is greater than that in NO^+

Solution:

Larger the magnitude of bond order, smaller will be the bond length.

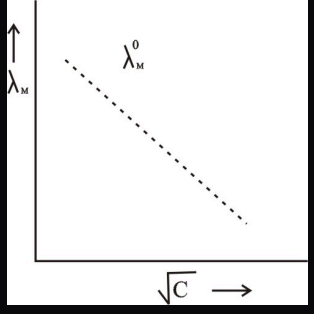
Q.27 Answer:

$\overset{(1)}{HCl}\ \overset{(2)}{NaCl}\ \overset{(3)}{NH_4OH}$

Solution:

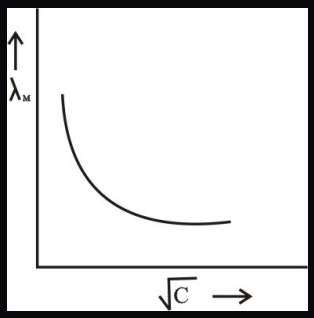
For strong electrolyte, curve given is straight line using Debye Huckel-Onsager equation.

$\lambda_m = \lambda_m^0 - k\sqrt{C}$



As HCl is the strongest. (1) curve matches to HCl and (2) curve to NaCl.

For weak electrolyte, when conc. decreases, α increases, the curve is given as



As $\mathrm{NH_4\,OH}$ is weak base so (3) curve matches to $\mathrm{NH_4\,OH}$

$\therefore$ option (c) is the correct answer.

Q.28 Answer:

$\mathrm{N_2(\,g) + 3H_2(\,g) \rightarrow 2NH_3(\,g)}$

Solution:

Entropy is the measure of randomness. As the randomness (disorder) increases, entropy increases, The order of entropy is

$$\underset{\text{prderticles}}{\text{Solid}} < \text{Liquid} < \underset{\text{(most disordered)}}{\text{Gas}}$$

Thus, in reaction $\mathrm{N_2(\,g) + 3H_2(\,g) \rightarrow 2NH_3(\,g)}$

Entropy decreases because number of gaseous molecules decreases.

Q.29 Answer:

$7.5 \times 10^{-3}\ \mathrm{N}$

Solution:

Mole of complex $= \frac{\mathrm{W}}{\mathrm{mol.wt.}} = \frac{1}{266.5}$

1 mole of complex produces 2 mole HCl

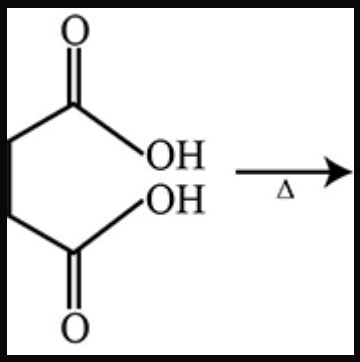
$\mathrm{[Cr\,(H_2O)_5\,Cl]\,Cl_2 \rightarrow Cr\,(H_2O)\,Cl]^{2+}\ 2Cl^-}$

Mole of HCl **formed** $= \frac{1}{266.5} \times 2$

$\mathrm{N_{HCl}} = \frac{\mathrm{n}}{\mathrm{V}} \times \mathrm{n-}$ **factor** $= \frac{2}{266.5 \times 1}$

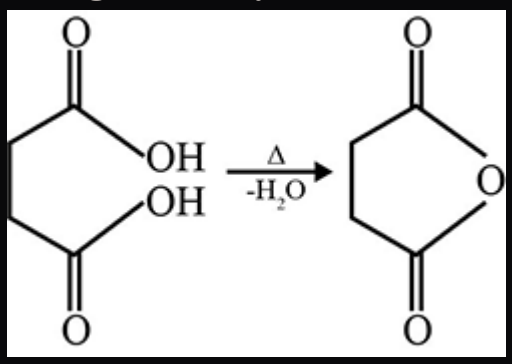
$= 7.5 \times 10^{-3}\ \mathrm{N}$

Q.30 Answer:



Solution:

Here this compound succinic acid undergoes dehydration to give succinic anhydride while rest other undergo decarboxylation.



Q.31 Answer:

$-3, +5$

Solution:

Ammonium nitrate (NH_4NO_3) exists as $NH_4^+NO_3^-$

In NH_4^+, N have oxidation state $x + 4 = +1$

$\Rightarrow x = -3$

In NO_3^-, N have oxidation state $x - 6 = -1$

$\Rightarrow x = +5$

Q.32 Answer:

II, IV

Solution:

Since in azulene (IV) two rings share a double bond, therefore, two electrons of this double bond can be counted towards each. As a result, each ring has 6π -electrons and hence azulene is aromatic. Pyrrole (II), the two double bonds + one lone pair of electrons on the N atom.

Q.33 Answer:

Na_2SO_3 and Na_2 S

Solution:

Here (A) & (B) are Na_2SO_3 and Na_2 S respectively.

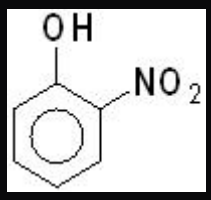
$$\underset{(X)}{Na_2SO_3} + 2HCl \rightarrow 2NaCl + \underset{(Y)}{SO_2} + H_2O$$

$K_2Cr_2O_7 + H_2SO_4 + 3SO_2 \rightarrow K_2SO_4 + Cr_2(SO_4)_3 + H_2O$

Green $\underset{(B)}{Na_2S} + 2HCl \rightarrow 2NaCl + \underset{(Z)}{H_2S}$

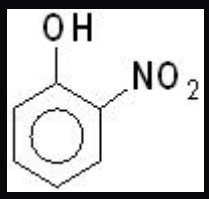
$Pb(CH_3COO)_2 + H_2S \rightarrow \underset{\text{Black}}{PbS} + 2CH_3COOH$

Q.34 Answer:



Solution:

Due to internal hydrogen bonding, hydrogen of $-OH$ is strongly bonded. So it can not behave as an acid and can not react with sodium bicarbonate. Ortho - nitro phenol is non polar and $NaHCO_3$ is polar. Like dissolves like.



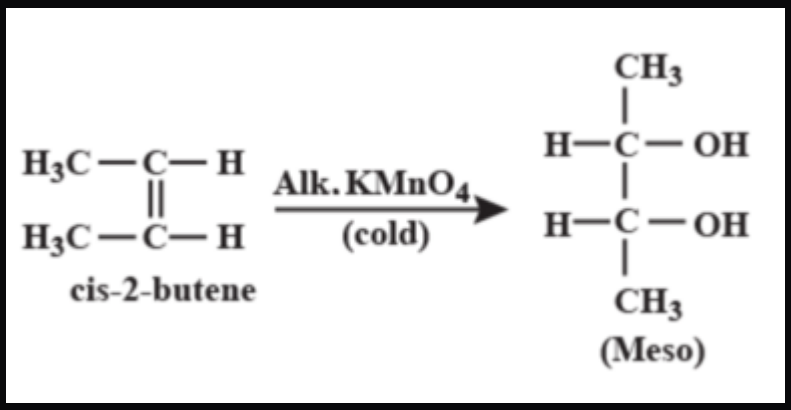
is not soluble in $NaHCO_3$

Q.35 Answer:

P is a the meso compound 2, 3-butanediol formed by syn-addition

Solution:

Cis-2-butene undergoes hydroxylation by syn-addition forming meso product.



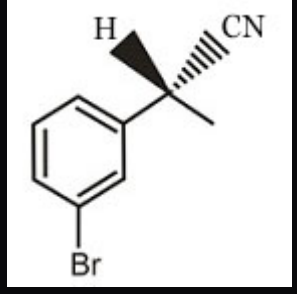
Q.36 Answer:

no blue colour formation

Solution:

There will be no blue colour formation because in the process of osmosis, solvent molecules migrate across the membrane and not solute molecules.

Q.37 Answer:



Solution:

S_N2 mechanism not possible at C-Br bond due to its double bond character caused by aromatic phenyl ring. But the reaction takes place C-Cl bond. The product is inverted compared to that of the substrate.

Q.38 Answer:

Rate $= \mathrm{k}\,[\mathrm{B_2}]$

Solution:

For the reaction $2\ \mathrm{A} + \mathrm{B_2} \rightleftharpoons 2\mathrm{AB}$,

Rate $\propto [\mathrm{A}]^{\mathrm{x}}\,[\mathrm{B_2}]^{\mathrm{y}}$. On substituting the given data, we get

From experimental

$1, 1.6 \times 10^{-4} \propto [0.50]^{\mathrm{x}}[0.50]^{\mathrm{y}}$ . . ...(i)

From experimental $2, 3.2 \times 10^{-4} \propto [0.50]^{x}[1.00]^{y}$ . . . (ii)

From experimental 3, $3.2 \times 10^{-4} \propto [1.00]^{x}[1.00]^{y}$ . . . (iii)

$1 = \left[\frac{1.00}{0.50}\right]^{\mathrm{x}}$

or $1 = 2^{\mathrm{x}}$ or $2^0 = 2^{\mathrm{x}}$ or $\mathrm{x} = 0$

Now, divide equation (ii) by equation (i) we get,

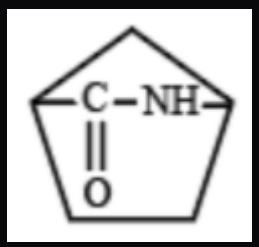
$2 = \left[\frac{1.00}{0.50}\right]^{\mathrm{y}}$

$2 = 2^{\mathrm{y}}$

$\mathrm{y} = 1$

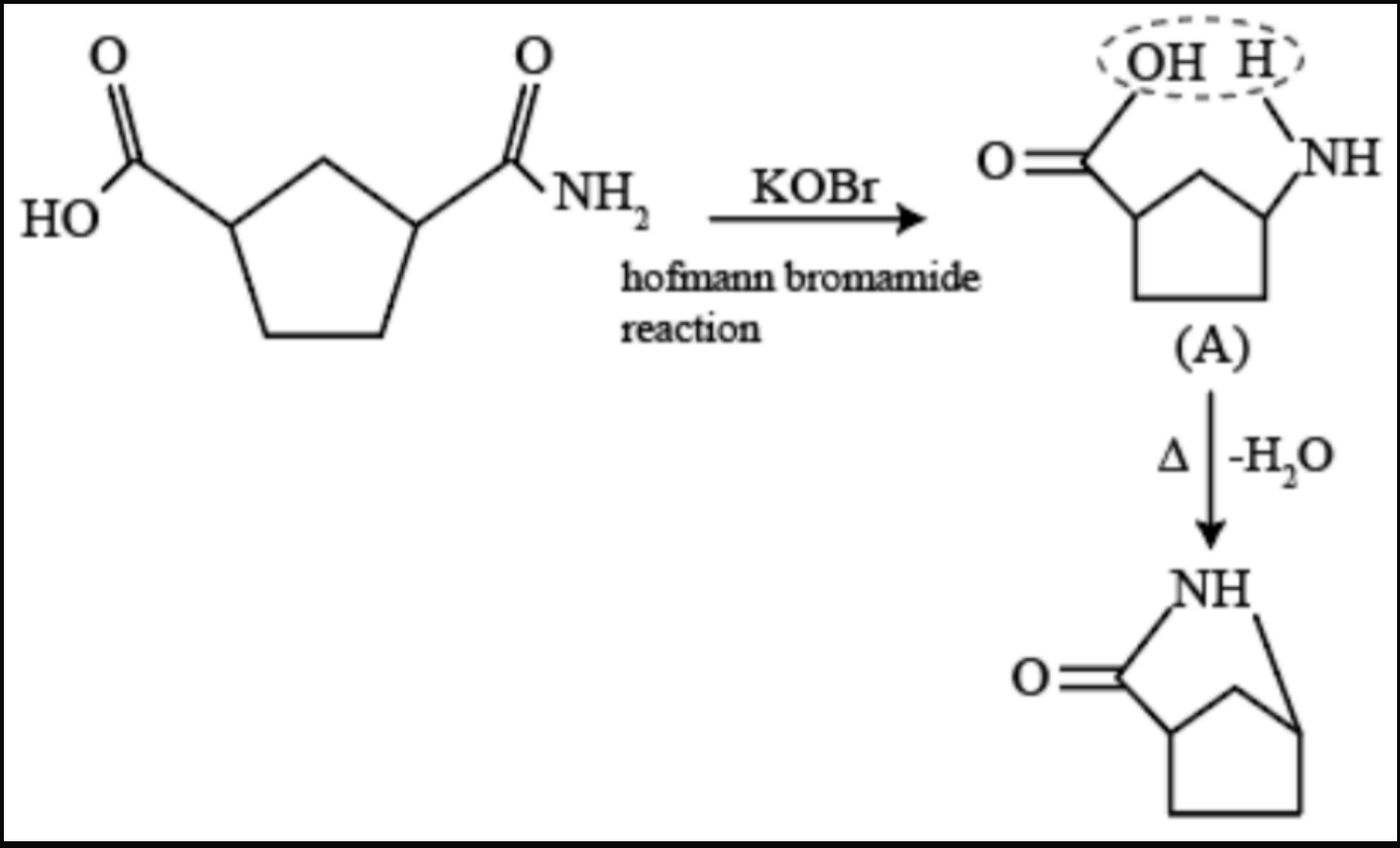
Thus rate equation is: Rate $= \mathrm{k}[\mathrm{A}]^0\,[\,\mathrm{B_2}]^1 = \mathrm{k}\,[\mathrm{B_2}]$.

Q.39 Answer:



Solution:

This conversion occurs as follows



Q.40 Answer:

$[Co(NH_3)_5SO_4]Br$ and $[Co(NH_3)_5Br]SO_4$

Solution:

Ionisation isomers in coordination chemistry is when complex compound having same molecule formula but gives different ions in aqueous solution on ionisation.

$[Co(NH_3)_5SO_4]Br$ and $[Co(NH_3)_5Br]SO_4$

$[CO\,(NH_3)_5\,SO_4]Br \overset{aq}{\rightarrow} [CO\,(NH_3)_5\,SO_4]^{+} + Br^{\ominus}$

$[CO\,(NH_3)_5\,Br]\,SO_4 \overset{aq}{\rightarrow} [CO\,(NH_3)_5\,Br]^{+2} + {SO_4}^{2-}$

Q.41 Answer:

$\frac{n^2h^2}{8k^{1/2}e\pi^2m^{3/2}}$

Solution:

Force $= \frac{d[U(r)]}{dr}$

$\therefore \frac{d[U(r)]}{dr} = \frac{4ke^2}{r^5}$

$\frac{4ke^2}{r^5} = \frac{mv^2}{r}$ **and** $mvr = \frac{nh}{2\pi}$

or $r = \frac{nh}{2\pi mv}$

$\Rightarrow \quad \frac{1}{r} = \frac{2\pi mv}{nh}$

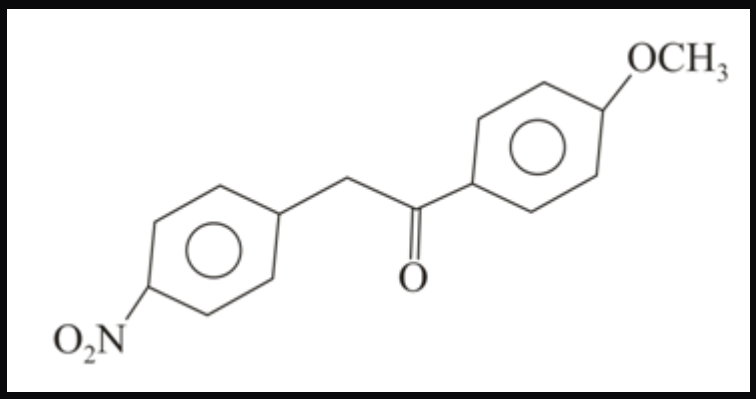
Now, $4ke^2 \times \frac{1}{r^5} = \frac{mv^2}{r}$

$4ke^2 \times \frac{16\pi^4m^4v^4}{n^4h^4} = mv^2$

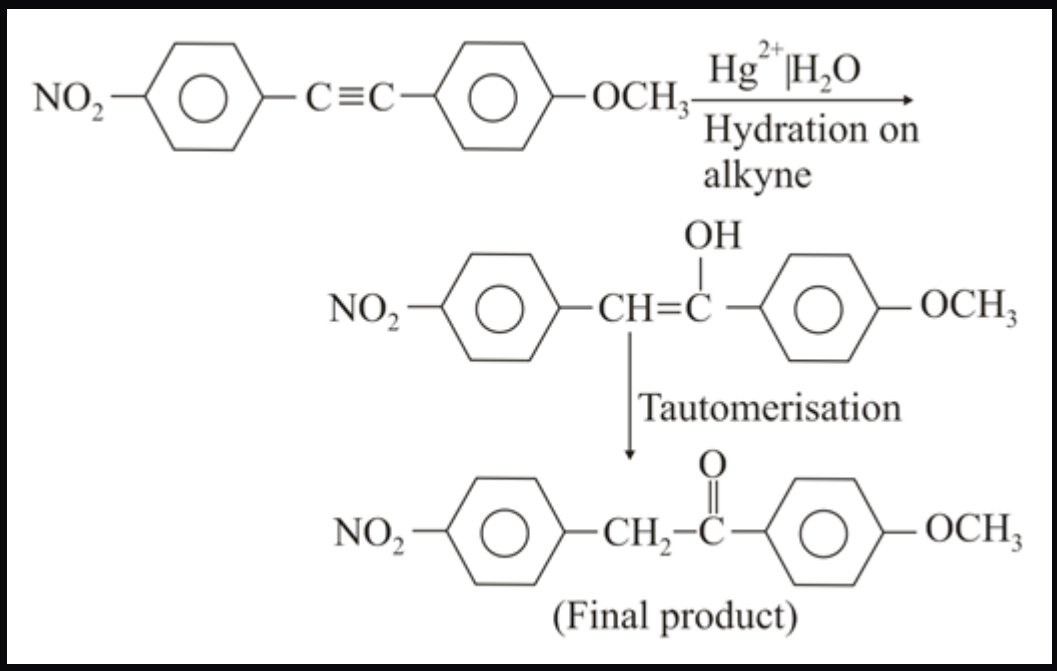
$v^2 = n^4h^4/64ke^2\pi^4m^3$

$v = \frac{n^2h^2}{8k^{1/2}e\pi^2m^{3/2}}$

Q.42 Answer:



Solution:

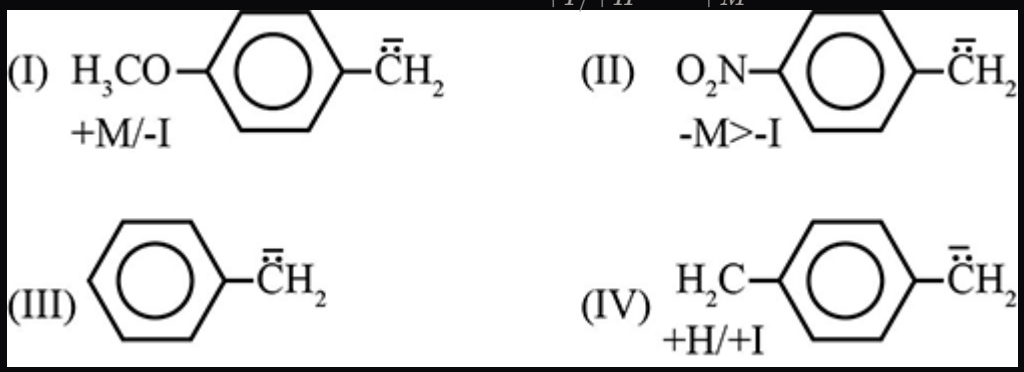


Q.43 Answer:

$II > III > IV > I$

Solution:

carbanion stability $\propto -M \propto -I \propto \frac{1}{+I/+H} \propto \frac{1}{+M}$



In (II), stability is increased as there is an electron withdrawing group, hence it stabilizes, but in (I) and (IV), it increases the electron density whereas in (III), there is no functional group attached for electron donation or withdrawal.

Q.44 Answer:

A – iii, B – i, C – ii, D – iv

Solution:

	Molecules	Hybridisation	Shape
(A)	PCl_5	sp^3d	**Trigonal bipyramidal**
(B)	OF_2	sp^3	**V-shape**
(C)	BCl_3	sp^2	**Trigonal planer**
(D)	NH_3	sp^3	**Trigonal pyramidal**

Q.45 Answer:

Rearrangement takes place

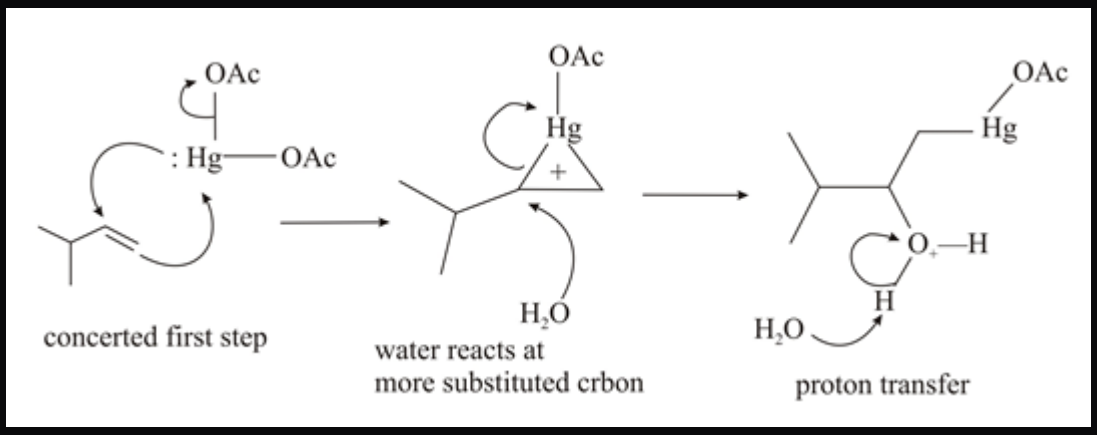
Solution:

Oxymercuration-Demercuration is exhibited by unsaturated alkenes or alkynes, where the unsaturated double bond undergoes addition reaction with water to give alcohols.

The reaction includes two major steps.

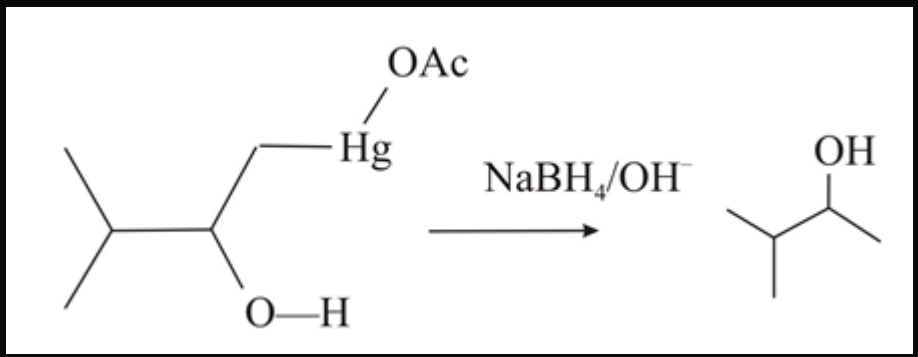
Oxymercuration:- Where mercuric acetate undergoes concerted reaction with alkenes to give a cyclic intermediate.

Lone pair present on water attacks the cyclic intermediate at highly substituted carbon thus there is no carbocation intermediate. So, rearrangement of carbocation is not possible.



.

During demercuration $NaBH_4$ reduces mercuric acetate to hydrogen.



.

Q.46 Answer:

—

Solution:

$\mathrm{Fe(s) \rightarrow Fe^{2+} + 2e^-; \Delta G_1^o}$

$\mathrm{2H^+ + 2e^- + \frac{1}{2}O_2 \rightarrow H_2O(l); \Delta G_2^o}$

$\mathrm{Fe(s) + 2H^+ + \frac{1}{2}O_2 \rightarrow Fe^{2+} + H_2O; \Delta G_3^o}$

Applying, $\Delta G_1^o + \Delta G_2^o = \Delta G_3^o$

$\Delta G_3^O = (-2\ \mathrm{F} \times 0.44) + (-2\ \mathrm{F} \times 1.23)$

$\Delta G_3^o = (-2 \times 96500 \times 0.44) + (-2 \times 96500 \times 1.23)$

$\Delta G_3^o = -322310\ \mathrm{J}$

$\therefore \Delta G_3^o = -322\ \mathrm{kJ}$

Q.47 Answer:

—

Solution:

Total pressure of mixture of gases A, B & $C = 10$ atm.
Partial pressure of $A =$ (No. of moles of $\mathrm{A} \times 10)/10$
or (No. of moles of $\mathrm{A} \times 10)/10 = 3$
Hence, no. of moles of $\mathrm{A} = 3$
$\therefore$ Partial pressure of $B =$ (No. of moles of $\mathrm{B} \times 10)/10$
(since partial pressure of $B = 1$ atm)
$\therefore$ No.of moles of $B = 1$
Now the no. of moles of $C = 10 - (3 + 1) = 6$;
1 mole of C weighs $= 2$ g.
$\therefore$ 6 mole of C will weigh $= 2 \times 6 = 12$

Q.48 Answer:

—

Solution:

The elements till atomic number 10 are hydrogen, helium, lithium, beryllium, boron, carbon, nitrogen, oxygen, fluorine and neon.
Electron affinity is defined as the amount of energy released when one mole of gaseous atoms accepts electrons to form a negative ion.
Electron gain enthalpy is said to be positive when energy is supplied to the gaseous atom for the addition of an electron and it is said to be negative when energy is released by the gaseous atom after the addition of an electron.
There are different factors that affect electron gain enthalpy.
(1) Electron gain enthalpy $\propto$ Nuclear charge
(2) Electron gain enthalpy $\propto \frac{1}{\text{Atomic size}}$

(3) Electron gain enthalpy also depends on the electronic configuration of atom. $\mathrm{He},\ \mathrm{Be},\ \mathrm{N}$ **and** Ne **have** $+\mathrm{ve}\ \ \mathrm{EGE}_1$**.**

Q.49 Answer:

—

Solution:

Total 8 **structural isomers are possible for** $\mathrm{C_5H_{11}Br}$**, out of which two isomer have Br at chiral carbon which can show stereochemical inversion.**

$\mathrm{CH_3CH_2CH_2CH_2(Br)CH_3}$ **and its enantiomer;** $\mathrm{(CH_3)_2CHCH(Br)CH_3}$ **and its enantiomer**

Q.50 Answer:

—

Solution:

At point of intersection

$[\mathrm{A_3}]_{\text{left}} = [\mathrm{A_2}]_{\text{formed}}$

$[\mathrm{A_3}]_{\text{left}} = [\mathrm{A_3}]_0\ \mathrm{e}^{-\mathrm{kt}}$ **.....(i)**

Concentration of $[\mathrm{A_3}]_{\text{used}}$

$= [\mathrm{A_3}]_0 - [\mathrm{A_3}]_0\ \mathrm{e}^{-\mathrm{kt}} = [\mathrm{A_3}]_0\left[1 - \mathrm{e}^{-\mathrm{kt}}\right]$

Concentration of $[\mathrm{A_2}]_{\text{formed}}$

$= \frac{3}{2}[\mathrm{A_3}]_{\text{used}} = \frac{3}{2}[\mathrm{A_3}]_0\left[1 - \mathrm{e}^{-\mathrm{kt}}\right]$

$= \frac{3}{2}[\mathrm{A_3}]_0 - \frac{3}{2}[\mathrm{A_3}]_0\mathrm{e}^{-\mathrm{kt}}$ **.....(ii)**

$[\mathrm{A_3}]_0\ \mathrm{e}^{-\mathrm{kt}} = \frac{3}{2}[\mathrm{A_3}]_0 - \frac{3}{2}[\mathrm{A_3}]_0\ \mathrm{e}^{-\mathrm{kt}}$

$\frac{5}{2}[\mathrm{A_3}]_0\ \mathrm{e}^{-\mathrm{kt}} = \frac{3}{2}[\mathrm{A_3}]_0$

$5\mathrm{e}^{-\mathrm{kt}} = 3$

$\ell\mathrm{n}\ 5 - \mathrm{kt} = \ell\mathrm{n}\ 3 \quad \therefore \quad \mathrm{t} = \frac{1}{\mathrm{k}}\ \ell\mathrm{n}\ \frac{5}{3}$

$\mathrm{t} = \frac{1}{\mathrm{k}}\ \ell\mathrm{n}\ \frac{\mathrm{Co}}{0.6\mathrm{Co}}$

i.e., 40 % Reaction.

Q.51 Answer:

$\frac{7}{2}$

Solution:

$$\lim_{n\to\infty}\prod_{r=3}^{n}\frac{(r+2)\left(r^2-2r+2^2\right)}{(r-2)\left(r^2+2r+2^2\right)}=\lim_{n\to\infty}\prod_{r=3}^{n}\frac{r+2}{r-2}\cdot\prod_{r=3}^{n}\frac{r^2-2r+2^2}{r^2+2r+2^2}$$

$$=\lim_{n\to\infty}\frac{5}{1}\cdot\frac{6}{2}\cdot\frac{7}{3}\cdot\frac{8}{4}\cdot\frac{9}{5}\cdots\frac{n+2}{n-2}\times\frac{7}{19}\cdot\frac{12}{28}\cdot\frac{19}{39}\cdots\frac{n^2-2n+4}{n^2+2n+4}$$

$$=\lim_{n\to\infty}\frac{(n-1)n(n+1)(n+2)}{1\cdot2\cdot3\cdot4}\cdot\frac{7\times12}{(n^2+3)\left(n^2+2n+4\right)}$$

$$=\lim_{n\to\infty}\frac{7}{2}\frac{\left(1-\frac{1}{n}\right)\left(1+\frac{1}{n}\right)\left(1+\frac{2}{n}\right)}{\left(1+\frac{3}{n^2}\right)\left(1+\frac{2}{n}+\frac{4}{n^2}\right)}=\frac{7}{2}.$$

Q.52 Answer:

$(4,-7,-9)$

Solution:

Required point $P=(r+2,-2r-3,-2r-5)$ and $r=2$

$\therefore P=(4,-7,-9)$

Q.53 Answer:

1

Solution:

Let $g(x)=f(f(x))$, given $f(x)$ is monotonic and differentiable

$\therefore \quad g'(x)=f'(f(x))\cdot f'(x)$, since $f(x)$ is monotonic $\Rightarrow f'(x)\cdot f'(f(x))>0$ for all $x\geq1$

Here, $\quad g(1)=f(f(1))$

$\therefore \quad g(1)=f(1)=1$

[given, $f(1)=1$]

Since, $g(x)$ is increasing for $x\geq1$.

$\therefore \quad g(x)\geq g(1)\Rightarrow\frac{1}{x^2-2x+2}\geq1$

$\Rightarrow\frac{1}{(x-1)^2+1}\geq1$, i.e. only possible when $x=1$.

$\therefore$ Number of solutions of $f(f(x))$ is 1.

Q.54 Answer:

$\log\left|x^2-y^2\right|+\tan^{-1}\frac{y}{x}=C$

Solution:

$$2(xdx-ydy)\left(x^2+y^2\right)=\left(x^2-y^2\right)(ydx-xdy)$$

$$\Rightarrow\frac{d\left(x^2-y^2\right)}{x^2-y^2}=\frac{-1}{1+\left(\frac{y}{x}\right)^2}\cdot d\left(\frac{y}{x}\right)$$

$$\Rightarrow\log\left|x^2-y^2\right|=-\tan^{-1}\frac{y}{x}+C$$

Q.55 Answer:

286

Solution:

Let E and W stand for one unit distance along positive x-direction and negative direction respectively.

$$EE \rightarrow \text{one way}$$

$$EEEW \rightarrow \binom{4}{1} = 4 \text{ ways}$$

$$EEEEWW \rightarrow \binom{6}{2} = 15 \text{ ways}$$

$$\text{EEEEEWWW} \rightarrow \binom{8}{3} = 56 \text{ ways}$$

$$\text{EEEEEEWWWW} \rightarrow \binom{10}{4} = 210 \text{ ways.}$$

The desired number of ways is $1 + 4 + 15 + 56 + 210 = 286$.

Q.56 Answer:

$\frac{5\sqrt{5}}{2}$

Solution:

Given triangle OAB is right angled triangle, right angled at 0.

$\therefore$ circum centre is $\left(1, \frac{11}{2}\right)$, orthocenter is $(0, 0)$.

Q.57 Answer:

$\sqrt{3}\mathrm{e}$

Solution:

$$\int \mathrm{e}^{\sin \mathrm{x}} \left(\frac{5 + \cos^2 \mathrm{x}}{(2 - \sin \mathrm{x})\sqrt{3 + \cos^2 \mathrm{x}}} \right) \cos \mathrm{x} d\mathrm{x}$$

$$\text{put } \sin \mathrm{x} = \mathrm{t}$$

$$\int \mathrm{e}^{\mathrm{t}} \left(\frac{6 - \mathrm{t}^2}{(2 - \mathrm{t})\sqrt{4 - \mathrm{t}^2}} \right) \mathrm{dt}$$

$$\int \mathrm{e}^{\mathrm{t}} \left(\frac{4 - \mathrm{t}^2}{(2 - \mathrm{t})\sqrt{4 - \mathrm{t}^2}} + \frac{2}{(2 - \mathrm{t})\sqrt{4 - \mathrm{t}^2}} \right) \mathrm{dt}$$

$$\int \mathrm{e}^{\mathrm{t}} \left(\sqrt{\frac{2 + \mathrm{t}}{2 - \mathrm{t}}} + \frac{2}{(2 - \mathrm{t})^{3/2}(2 + \mathrm{t})^{1/2}} \right) \mathrm{dt}$$

If $g(t)=\sqrt{\frac{2+t}{2-t}}, g'(t)=\frac{2}{(2-t)^{3/2}\left(2+t^2\right)^{1/2}}$

$e^t \sqrt{\frac{2+t}{2-t}}+c$

$f(x)=e^{\sin x} \sqrt{\frac{2+\sin x}{2-\sin x}}+c$

$f(0)=1 \Rightarrow c=0$

$f(x)=e^{\sin x} \sqrt{\frac{2+\sin x}{2-\sin x}}$

$f\left(\frac{\pi}{2}\right)=\sqrt{3} \mathrm{e}$

Q.58 Answer:

$x^2+2x-1=0$

Solution:

$\vec{b}-2\vec{c}=\alpha\vec{a} \Rightarrow \vec{b}=2\vec{c}+\alpha\vec{a}$

$\Rightarrow|\vec{b}|^2=|2\vec{c}+\alpha\vec{a}|^2=4|\vec{c}|^2+\alpha^2|\vec{a}|^2+4\alpha\vec{a}\cdot\vec{c}$

$\Rightarrow 5=4+\alpha^2+(4\alpha)\left(\frac{1}{2}\right)$

$\alpha^2+2\alpha-1=0$

Q.59 Answer:

1

Solution:

Given $A-1=xA2+yA+zI$

$$A^{-1}=I=xA^3+yA^2+zA \quad \ldots(1)$$

$$A^2=\left[\begin{array}{lll}2 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 2\end{array}\right]\left[\begin{array}{lll}2 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 2\end{array}\right]\left[\begin{array}{lll}6 & 4 & 5\\ 4 & 3 & 4\\ 5 & 4 & 6\end{array}\right]$$

$$A^3=\left[\begin{array}{lll}6 & 4 & 5\\ 4 & 3 & 4\\ 5 & 4 & 6\end{array}\right]\left[\begin{array}{lll}2 & 1 & 1\\ 1 & 1 & 1\\ 1 & 1 & 2\end{array}\right]\left[\begin{array}{lll}21 & 15 & 20\\ 15 & 11 & 15\\ 20 & 15 & 21\end{array}\right]$$

Now using $\mathrm{xA}^3+y\mathrm{A}^2+\mathrm{zA}=I$

$$\left[\begin{array}{lll}21x & 15x & 20x\\ 15x & 11x & 15x\\ 20x & 15x & 21x\end{array}\right]+\left[\begin{array}{lll}6y & 4y & 5y\\ 4y & 3y & 4y\\ 5y & 4y & 6y\end{array}\right]+\left[\begin{array}{lll}2z & z & z\\ z & z & z\\ z & z & 2z\end{array}\right]=\left[\begin{array}{lll}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{array}\right]$$

Comparing the diagonal elements

$21x + 6y + 2z = 1 \quad \ldots(1)$

$11x + 3y + z = 1 \quad \ldots(2)$

$21x + 6y + 2z = 1 \quad \ldots(3)$

$2 \times (2) - (1)$ gives $\quad x = 1$

$3y + z = -10 \quad \ldots(4)$

Comparing, a12

$15x + 4y + z = 0$

$4y + z = -15 \quad \ldots(5)$

for (4) & $(5) \quad y = -5$ and $z = 5$

$x + y + z = 1 + (-5) + 5 = 1$

Q.60 Answer:

0

Solution:

$$A_1 = \begin{pmatrix} n \\ r-1 \end{pmatrix}, A_2 = \begin{pmatrix} n \\ r \end{pmatrix}, A_3 = \begin{pmatrix} n \\ r+1 \end{pmatrix}, A_4 = \begin{pmatrix} n \\ r+2 \end{pmatrix}.$$

$$\frac{A_1}{A_1 + A_2} = \frac{1}{1 + \frac{A_2}{A_1}} = \frac{1}{1 + \left\{ \begin{pmatrix} n \\ r \end{pmatrix} / \begin{pmatrix} n \\ r-1 \end{pmatrix} \right\}}$$

$$= \frac{1}{1 + \frac{n-r+1}{r}} = \frac{r}{n+1}, \frac{A_2}{A_2 + A_3} = \frac{r+1}{n+1}, \frac{A_3}{A_3 + A_4} = \frac{r+2}{n+1}$$

$$\therefore \frac{A_1}{A_1 + A_2} + \frac{A_3}{A_3 + A_4} - \frac{2A_2}{A_2 + A_3} = \frac{r}{n+1} + \frac{r+2}{n+1} - \frac{2(r+1)}{n+1} = 0.$$

Q.61 Answer:

-6

Solution:

adding these equation, we get

$$-2 = \frac{2k}{3} - 5 + \frac{k^2}{2}$$

i.e. $3k^2 + 4k - 18 = 0$

$$k_1 k_2 = \frac{-18}{3} = -6$$

Q.62 Answer:

an equivalence relation

Solution:

The given relation may be written in set builder form as

$\mathrm{R} = \{(\mathrm{a}, \mathrm{b}) : \mathrm{a} - \mathrm{b}$ **divided by** n **and** $\mathrm{a}, \mathrm{b} \in \mathrm{Z}\}$

As $\mathrm{a} - \mathrm{a} = 0$ **and 0 divided by** n

$\therefore (a, a) \in R$

$\therefore R$ is reflexive

Let $a, b \in Z$ such that $(a, b) \in R$

Then $(a, b) \in R \Rightarrow a - b$ divided by n.

$a - b = nk$ for some integer k

$\Rightarrow b - a = n(-k)$

$\therefore (a, b) \in R \Rightarrow (b, a) \in R$

$\therefore R$ is symmetric

Now, (a, b), (b, c) $\in R$

Now, $a - b = nc_1$ and $b - c = nc_2$ for some integers c_1 and c_2.

$\therefore (a - b) + (b - c) = n(c_1 + c_2)$

$\Rightarrow a - c = nk$, where $k = c_1 + c_2$, an integer.

$\Rightarrow (a, c) \in R$.

$\therefore (a, b), (b, c) \in R \Rightarrow (a, c) \in R$

$\therefore$ R is transitive and hence R is an equivalence relation.

Q.63 Answer:

-5

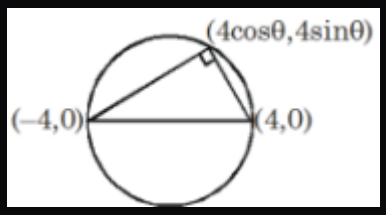
Solution:

$$\alpha + \beta + \gamma + \delta = 0; \alpha\beta\gamma\delta = -5$$
$$(\alpha + \beta + \gamma)(\alpha + \beta + \delta)(\beta + \gamma + \delta)(\alpha + \gamma + \delta) = \alpha\beta\gamma\delta = -5$$

Q.64 Answer:

7

Solution:



$\text{Area } = \frac{1}{2} \times 8 \times 4 \sin\theta = |16 \sin\theta|$

If area is integer then $\sin\theta$ can be equal to $\frac{1}{16}, \frac{2}{16}, \frac{3}{16} \cdots \frac{15}{16}$

15 in all four quadrants and 2 are on aixs $\left[\frac{62}{8}\right] = 7$

Q.65 Answer:

Both I, II are true

Solution:

$$P(A \cup B) \geq \frac{3}{4} \Rightarrow P(A) + P(B) \geq \frac{3}{4} + P(A \cap B)$$
$$\geq \frac{3}{4} + \frac{1}{8} \geq \frac{7}{8}$$
$$P(A \cup B) \leq 1 \Rightarrow P(A) + P(B) \leq 1 + \frac{3}{8} \leq \frac{4}{8}$$

Q.66 Answer:

0

Solution:

$\because$ 1 lies between roots of $y^2 - my + 1 = 0$
$\Rightarrow \mathrm{f}(1) < 0 \Rightarrow \mathrm{m} > 2$
Let $z = \frac{4|x|}{x^2+16} = \frac{4t}{t^2+16} \Rightarrow t = |x|$
$\because zt^2 - 4\mathrm{t} + 16\mathrm{z} = 0$
$\because \mathrm{t} \geq 0$
$0 \leq \mathrm{z} \leq \frac{1}{2}$
$\Rightarrow [z^{\mathrm{m}}] = 0 \Rightarrow \mathrm{m} > 2$

Q.67 Answer:

$(0, 108]$

Solution:

$\frac{a+\frac{b}{2}+\frac{b}{2}+\frac{c}{3}+\frac{c}{3}+\frac{c}{3}}{6} \geq \sqrt[6]{a\left(\frac{b}{2}\right)^2\left(\frac{c}{3}\right)^3}$

Q.68 Answer:

1

Solution:

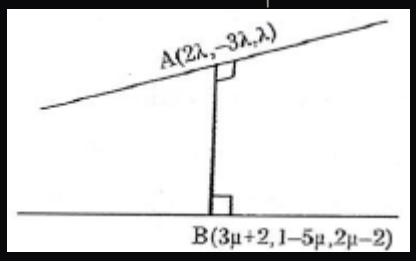
Put $\mathrm{z} = \mathrm{x} + \mathrm{iy}$
$(x + iy) + i(|z|) = i(x - iy) + 1$
$\Rightarrow x = y + 1 \& y + |z| = x$
$\Rightarrow x - y = 1 \& x - y = |z|$
$\Rightarrow |z| = 1$

Q.69 Answer:

$\frac{(x-21)}{3} = \frac{\left(y+\frac{92}{3}\right)}{3} = \frac{\left(z-\frac{32}{3}\right)}{3}$

Solution:

Line has direction $\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -3 & 1 \\ 3 & -5 & 2 \end{vmatrix} = -\hat{i} - \hat{j} - \hat{k}$



$2\lambda - 3\mu - 2 = 5\mu - 3\lambda - 1 = \lambda + 2 - 2\mu$

Solving $\lambda = \frac{31}{3}$ & $\mu = \frac{19}{3}$

Point B is $\left(21, -\frac{92}{3}, \frac{32}{3}\right)$

Q.70 Answer:

one

Solution:

Given, $f(x+y) = f(x) + f(y) - xy - 1, \forall x, y \in R$

$\therefore f(x+1) = f(x) + f(1) - x - 1[\text{ putting } y = 1]$

$\Rightarrow f(x+1) = f(x) - x[\because f(1) = 1]$

$\therefore f(n+1) = f(n) - n < f(n)$

$\Rightarrow f(n+1) < f(n)$

So, $f(n) < f(n-1) < f(n-2) < \ldots < f(3) < f(2) < f(1) = 1$

$\therefore f(n) = n$ holds only for $n = 1$

Q.71 Answer:

—

Solution:

Adding mean to the observations does not change the value of $\sum (x_i - \bar{x})^2$, hence,

$$\frac{\left(\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n}\right)}{\left(\frac{\sum_{i=1}^{n+1}(x_i - \bar{x})^2}{n+1}\right)} = \frac{78}{72}$$

$$\Rightarrow \frac{n+1}{n} = \frac{78}{72} \Rightarrow n = 12$$

Q.72 Answer:

—

Solution:

Ellipse $16x^2 + 25y^2 = 400$

$\Rightarrow \frac{x^2}{25} + \frac{y^2}{16} = 1$

Let the points on ellipse be $\mathrm{P}\left(5\cos\theta_1, 4\sin\theta_1\right)$ and $Q\left(5\cos\theta_2, 4\sin\theta_2\right)$. Circle described on PQ

as diameter touches x-axis $(3,0)$. Then

$5\left(\frac{\cos\theta_1+\cos\theta_2}{2}\right)=3$ and $4\left(\frac{\sin\theta_1+\sin\theta_2}{2}\right)=r$

$\Rightarrow \frac{4}{5}\tan\left(\frac{\theta_1+\theta_2}{2}\right)=\frac{r}{3}$

$\Rightarrow \tan\left(\frac{\theta_1+\theta_2}{2}\right)=\frac{5r}{12}$

$\Rightarrow$ Slope of $PQ=-\frac{4}{5}\cot\frac{\theta_1+\theta_2}{2}$

$=-\frac{48}{25r}=-1$

Therefore, $1/|\mathrm{m}|=1$

Q.73 Answer:

—

Solution:

A is orthogonal matrix

$a^2+b^2+c^2=1 \quad ab+bc+ca=0$

$a^3+b^3+c^3=(a+b+c)\left(a^2+b^2+c^2-ab-bc-ca\right)+3abc$

$=(a+b+c)(1)+3$

$(a+b+c)^2=a^2+b^2+c^2-2(ab+bc+ca)$

$=1$

$a+b+c=1$

$\Rightarrow a^3+b^3+c^3=4$

Q.74 Answer:

—

Solution:

$$\mathrm{I}=\int_{\frac{\pi}{4}}^{\frac{\pi}{3}}\frac{\left(\sin^3\theta-\cos^3\theta-\cos^2\theta\right)\left(\sin\theta+\cos\theta+\cos^2\theta\right)^{2007}}{(\sin\theta)^{2009}(\cos\theta)^{2009}}\,\mathrm{d}\theta$$

$$=\int_{\frac{\pi}{4}}^{\frac{\pi}{3}}\left(\tan\theta\sec\theta-\cot\theta\,\mathrm{cosec}\,\theta-\mathrm{cosec}^2\,\theta\right)\left(\sec\theta+\mathrm{cosec}\,\theta+\cot\theta\right)^{2007}\,\mathrm{d}\theta$$

$\sec\theta+\mathrm{cosec}\,\theta+\cot\theta=\mathrm{t}$

$\left(\sec\theta\tan\theta-\cot\theta\,\mathrm{cosec}\,\theta-\sec^2\theta\right)\mathrm{d}\theta=\mathrm{dt}$

$$I=\int_{\sqrt{8}+1}^{2+\sqrt{3}}\mathrm{t}^{2007}\mathrm{dt}=\left(\frac{\mathrm{t}^{2008}}{2008}\right)_{\sqrt{8}+1}^{2+\sqrt{3}}$$

$$= \frac{(2+\sqrt{3})^{2008} - (\sqrt{8}+1)^{2008}}{2008}$$

$A = 2, b = 3, C = 8, d = 2008$

$a + b + C + d = 2 + 3 + 8 + 2008 = 2021$

Q.75 Answer:

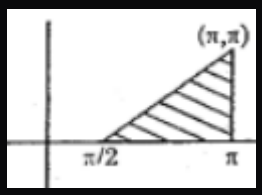
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Solution:

$$y = \left|\frac{\pi}{2} - \sin^{-1}(\sin x)\right| + \left|\frac{\pi}{2} - \cos^{-1}(\cos x)\right|$$

$$y = \left|\frac{\pi}{2} - (\pi - x)\right| + \left|\frac{\pi}{2} - x\right|$$

$$y = \left|x - \frac{\pi}{2}\right| + \left|\frac{\pi}{2} - x\right| = 2x - \pi \quad \forall x \in \left[\frac{\pi}{2}, \pi\right]$$



Area $= \frac{1}{2}\left(\frac{\pi}{2}\right)\pi = \frac{\pi^2}{4}$